

MICRO X Rev A — printed part reference drawings

Actuated chibi T-rex; 14 policy joints + jaw. Millimetres. Assembly frame: X forward, Y left, Z up, floor Z=0.

21 printed parts (421.6 g solid PLA equivalent) + 18 purchased items (404.0 g catalogue).

Overall views generated from the actual meshes; use STEP for exact geometry. Not production GD&T; prototype fit and fastener checks are still required.

Printed parts

left_yaw2roll	link left_yaw2roll	30.0 x 26.0 x 36.0 mm	5.7 g
left_hip	link left_hip	36.0 x 70.3 x 50.6 mm	18.5 g
left_thigh	link left_thigh	47.8 x 40.2 x 63.1 mm	18.2 g
left_shin	link left_shin	50.7 x 33.9 x 60.3 mm	14.9 g
left_foot	link left_foot	90.0 x 50.4 x 49.1 mm	35.5 g
right_yaw2roll	link right_yaw2roll	30.0 x 26.0 x 36.0 mm	5.7 g
right_hip	link right_hip	36.0 x 70.3 x 50.6 mm	18.5 g
right_thigh	link right_thigh	47.8 x 40.2 x 63.1 mm	18.2 g
right_shin	link right_shin	50.7 x 33.9 x 60.3 mm	14.9 g
right_foot	link right_foot	90.0 x 50.4 x 49.1 mm	35.5 g
chassis	link trunk	138.0 x 58.4 x 64.0 mm	43.3 g
torso_shell	link trunk	106.6 x 77.9 x 52.5 mm	26.8 g
tail_cover	link trunk	78.0 x 66.0 x 48.0 mm	32.7 g
neck_link	link neck	40.1 x 35.5 x 72.1 mm	14.1 g
head_base	link head_base	24.0 x 30.5 x 31.7 mm	3.6 g
head_yoke	link head_yoke	55.0 x 41.5 x 31.5 mm	21.1 g
head_frame	link head	91.5 x 34.0 x 42.0 mm	16.9 g
skull	link head	122.0 x 79.9 x 61.0 mm	47.9 g
eye_left	link head	30.0 x 26.0 x 30.0 mm	8.1 g
eye_right	link head	30.0 x 26.0 x 30.0 mm	8.1 g
jaw_beak	link jaw	36.1 x 58.4 x 29.0 mm	13.6 g

Purchased items (envelopes)

servo_left_hip_yaw	actuator	link trunk	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_left_hip_roll	actuator	link left_yaw2roll	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_left_hip_pitch	actuator	link left_thigh	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_left_knee	actuator	link left_shin	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_left_ankle	actuator	link left_shin	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_neck_pitch	actuator	link trunk	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_head_pitch	actuator	link neck	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_head_yaw	actuator	link head_yoke	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_head_roll	actuator	link head_yoke	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_right_hip_yaw	actuator	link trunk	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_right_hip_roll	actuator	link right_yaw2roll	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_right_hip_pitch	actuator	link right_thigh	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_right_knee	actuator	link right_shin	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_right_ankle	actuator	link right_shin	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
servo_jaw	actuator	link head	18.0 g	XL330-class smart servo envelope; verify drawing before ordering
battery_np_f550	battery	link trunk	100.0 g	NP-F550 class 7.4 V 2600 mAh removable pack; dimensions 70.8 x 38.4 x
compute_board	electronics	link trunk	30.0 g	Vertical carrier for a compute module (50 x 36 usable outline); PCB pl
camera_module_3	camera	link head	4.0 g	Raspberry Pi Camera Module 3 Standard class, 25 x 24 mm board

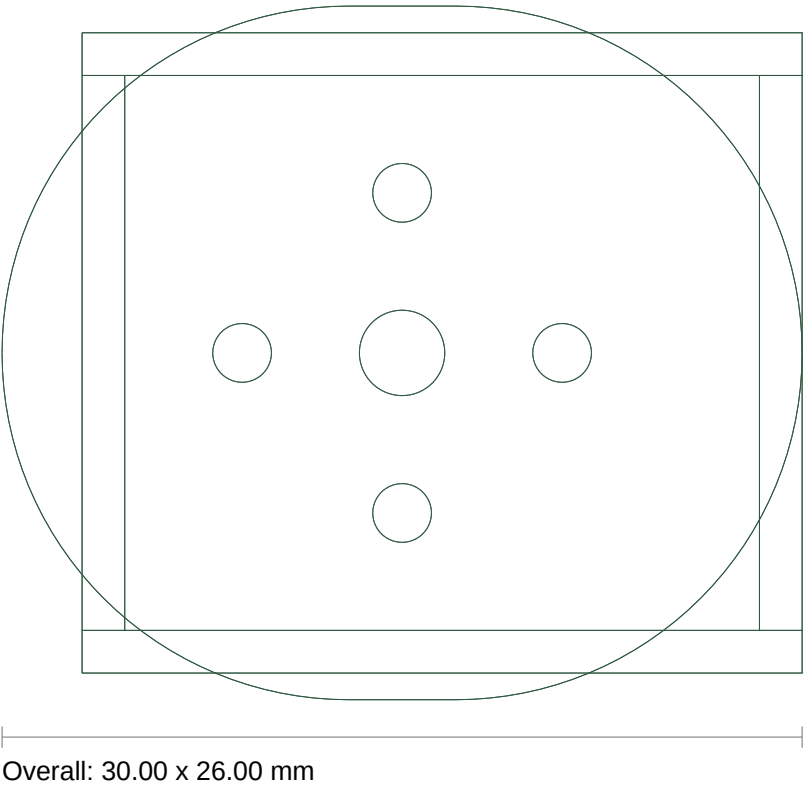
MICRO X Rev A / left_yaw2roll

link: left_yaw2roll | millimetres | original design, prototype reference, not production released

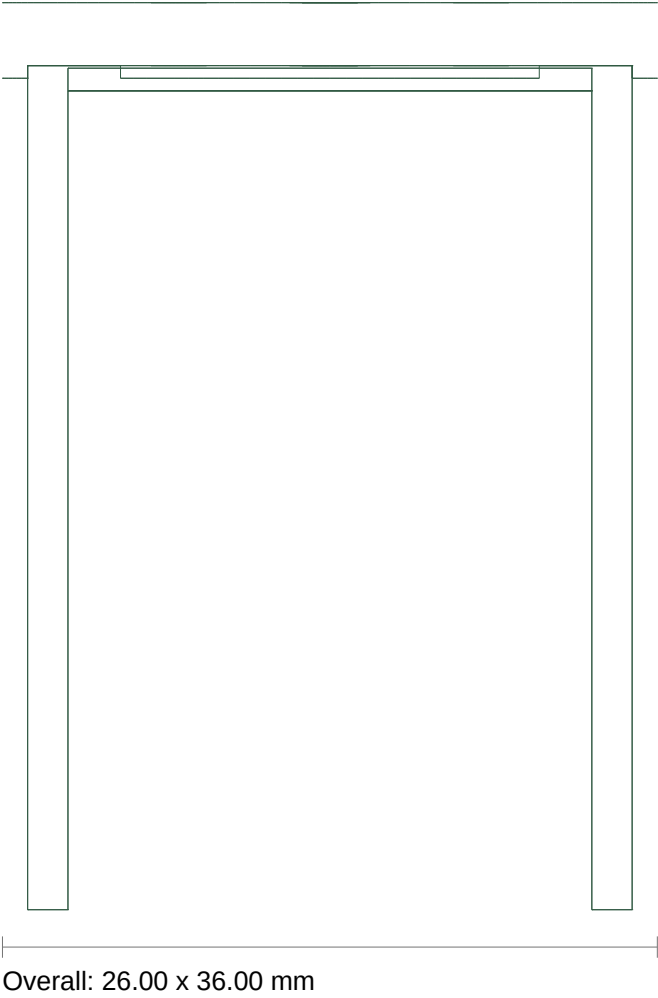
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 30.0 / 26.0 / 36.0 mm

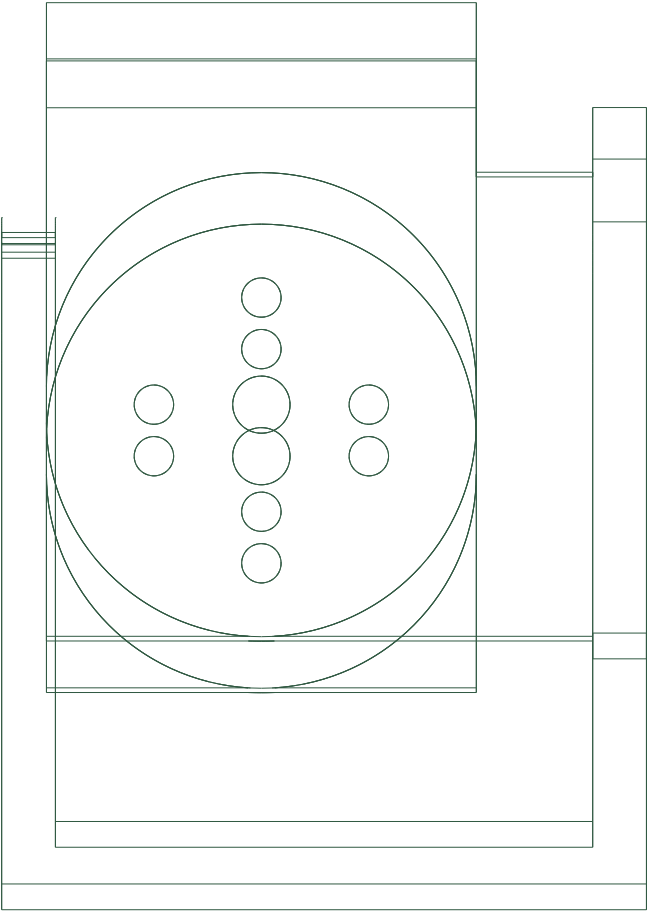
Material volume: 4568.0 mm3 | PLA solid-equivalent: 5.66 g (slicer mass will be lower)

Hip yaw horn plate with hip roll servo cradle; servo drops in from below and is retained by the hip bracket horn/idler bolts. M2 case screws to be confirmed.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / left_hip

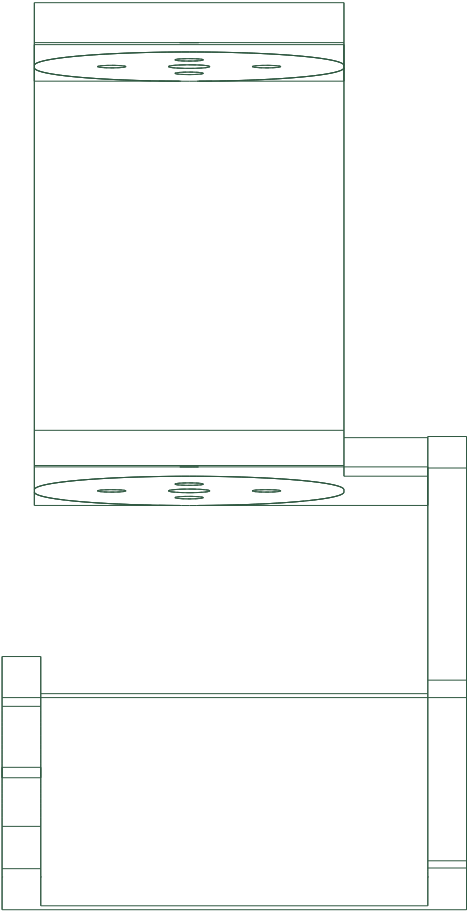
link: left_hip | millimetres | original design, prototype reference, not production released

SIDE X-Z



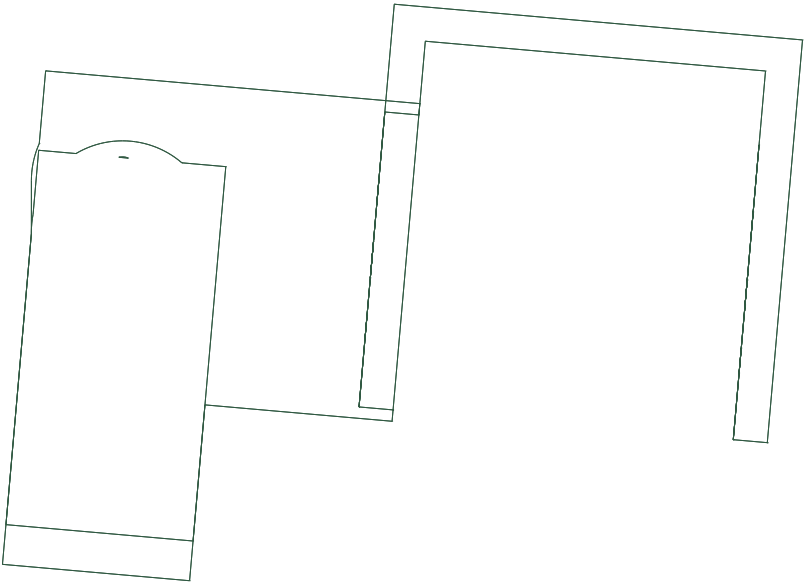
Overall: 36.00 x 50.63 mm

TOP X-Y



Overall: 36.00 x 70.28 mm

FRONT Y-Z



Overall: 70.28 x 50.63 mm

Assembly-frame extents X/Y/Z: 36.0 / 70.28 / 50.63 mm

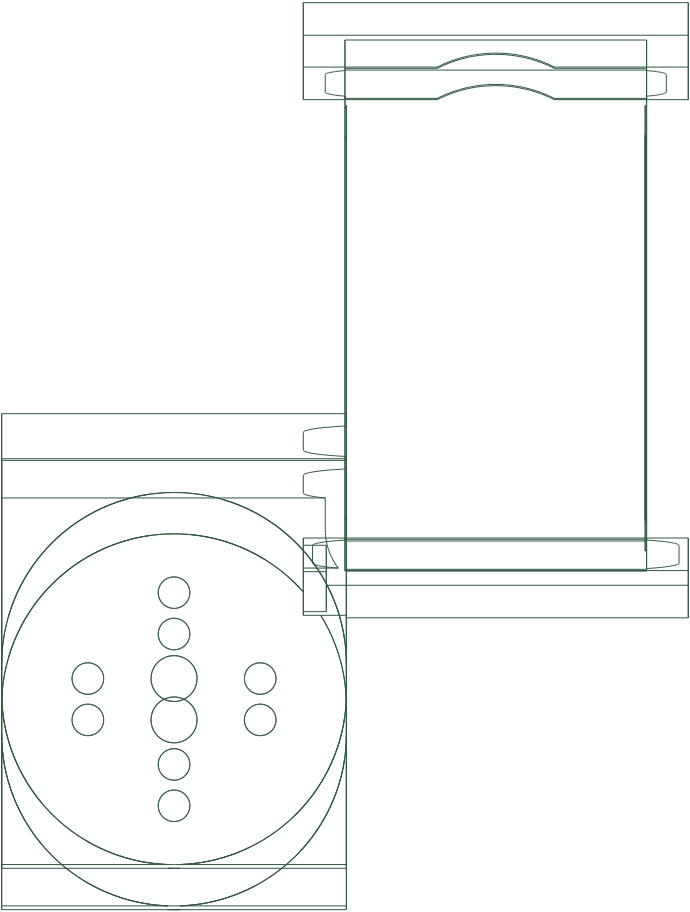
Material volume: 14895.1 mm3 | PLA solid-equivalent: 18.47 g (slicer mass will be lower)

Hip bracket: roll horn/idler plates with an under-bar, pitch horn/idler plates and a tie above the pitch servo cap. Pitch servo body lives in the thigh.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / left_thigh

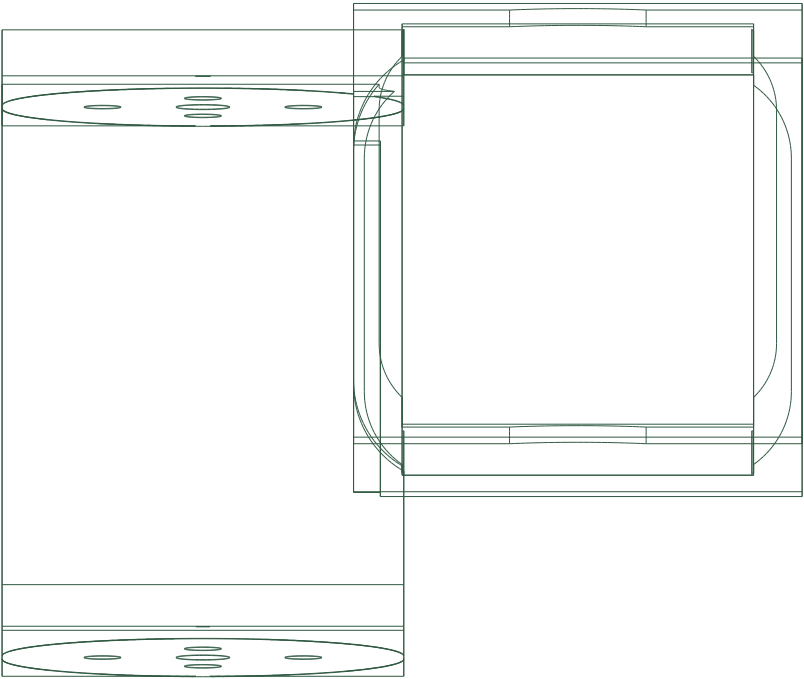
link: left_thigh | millimetres | original design, prototype reference, not production released

SIDE X-Z



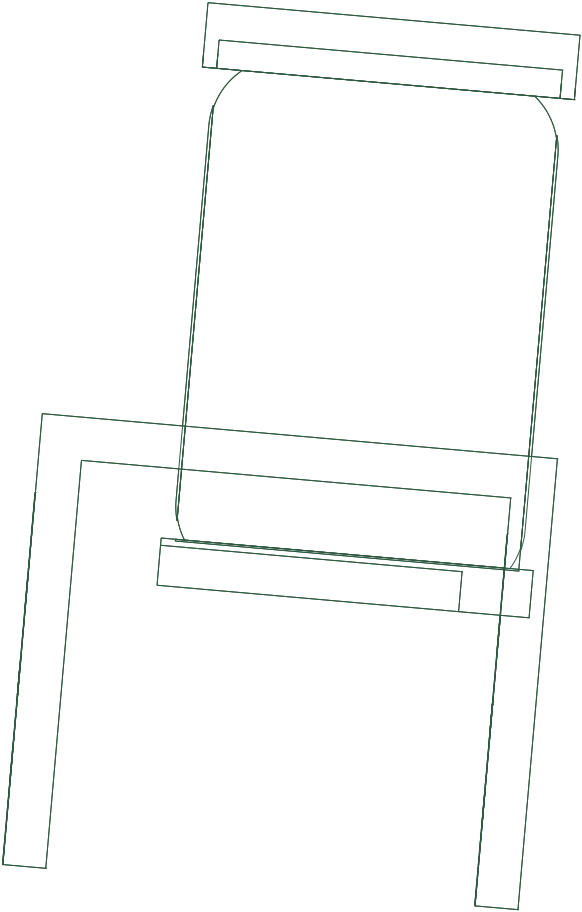
Overall: 47.80 x 63.13 mm

TOP X-Y



Overall: 47.80 x 40.21 mm

FRONT Y-Z



Overall: 40.21 x 63.13 mm

Assembly-frame extents X/Y/Z: 47.8 / 40.21 / 63.13 mm

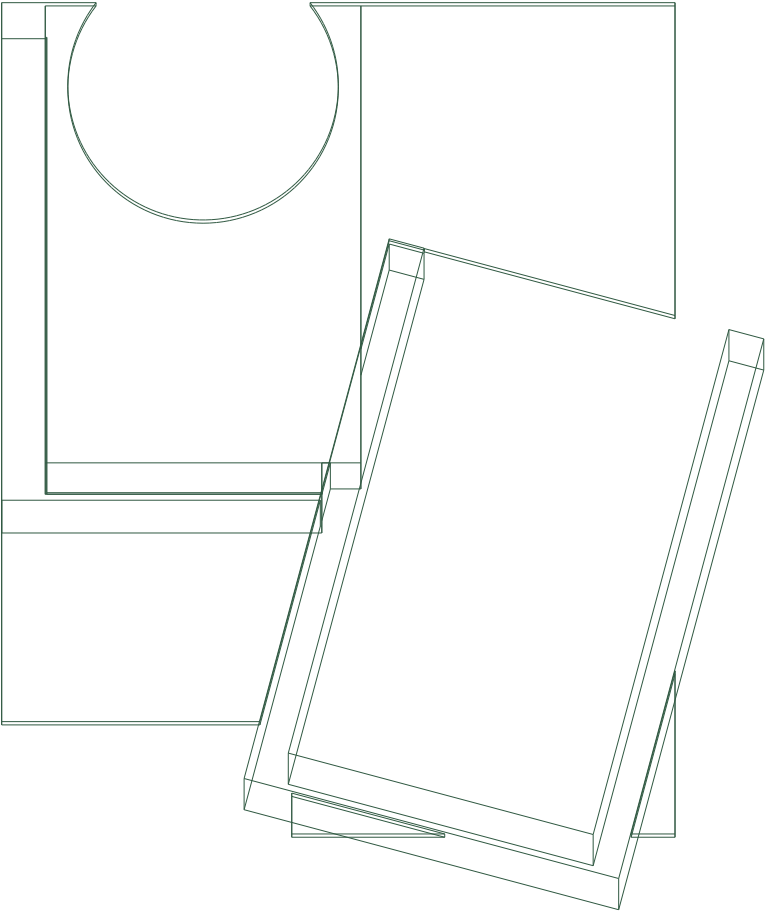
Material volume: 14711.7 mm3 | PLA solid-equivalent: 18.24 g (slicer mass will be lower)

Thigh: houses the hip pitch servo, carries the knee horn and idler plates with a tie above the knee servo.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / left_shin

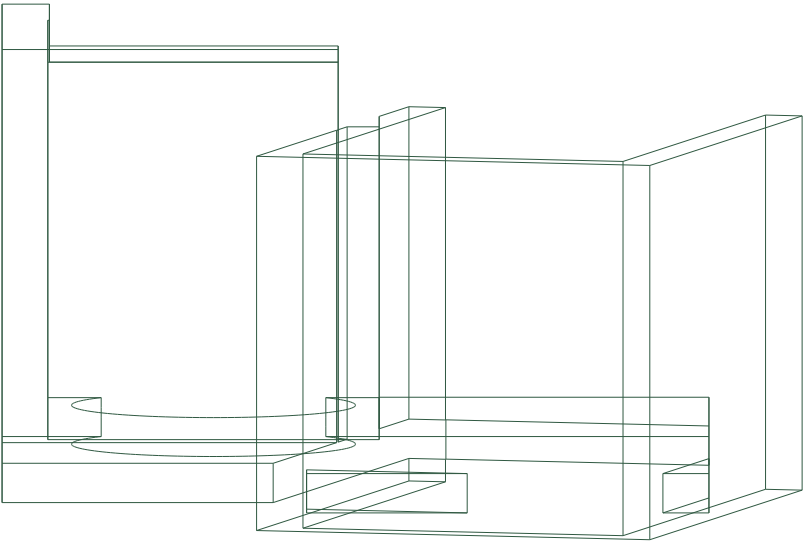
link: left_shin | millimetres | original design, prototype reference, not production released

SIDE X-Z



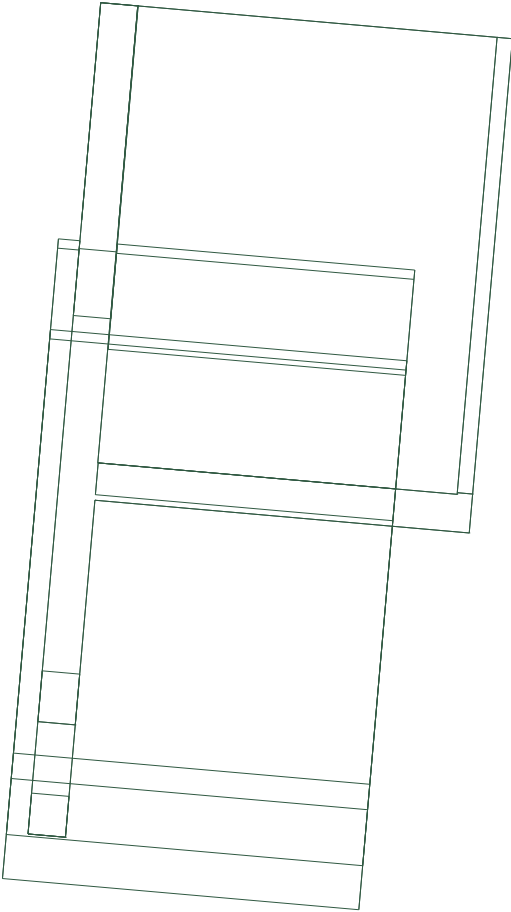
Overall: 50.71 x 60.33 mm

TOP X-Y



Overall: 50.71 x 33.93 mm

FRONT Y-Z



Overall: 33.93 x 60.33 mm

Assembly-frame extents X/Y/Z: 50.71 / 33.93 / 60.33 mm

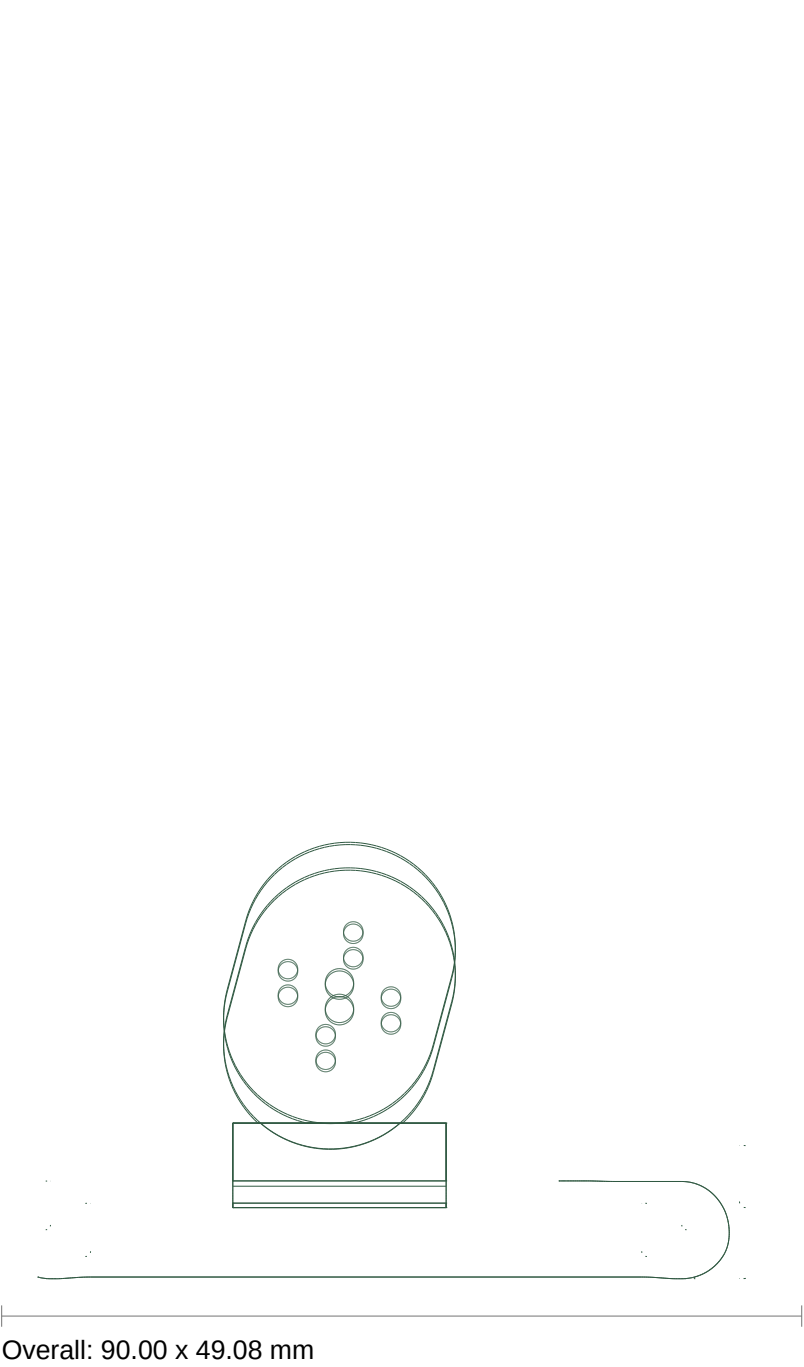
Material volume: 11986.8 mm3 | PLA solid-equivalent: 14.86 g (slicer mass will be lower)

Shin: spine plate with knee servo bay above an upright ankle servo cradle; both servos retained by the neighbouring clevis plates.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

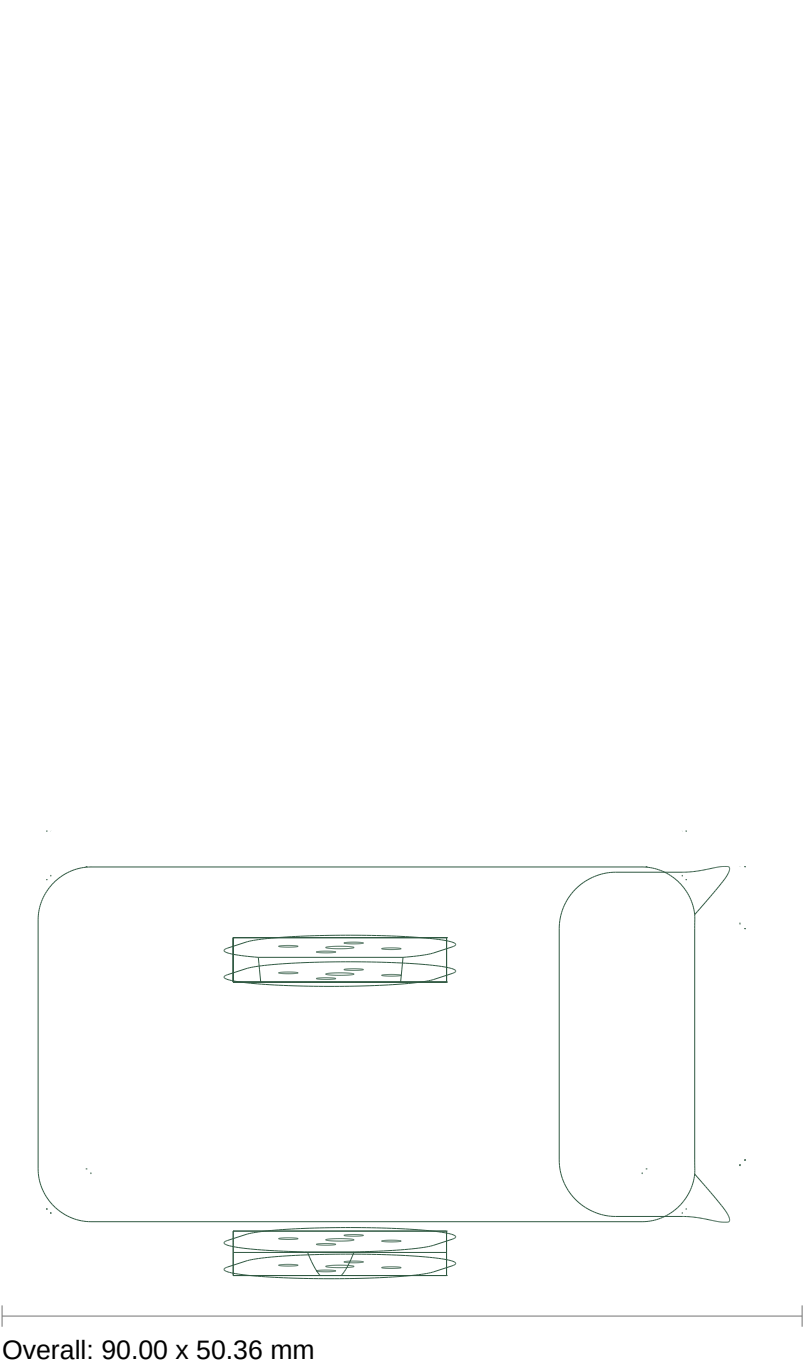
MICRO X Rev A / left_foot

link: left_foot | millimetres | original design, prototype reference, not production released

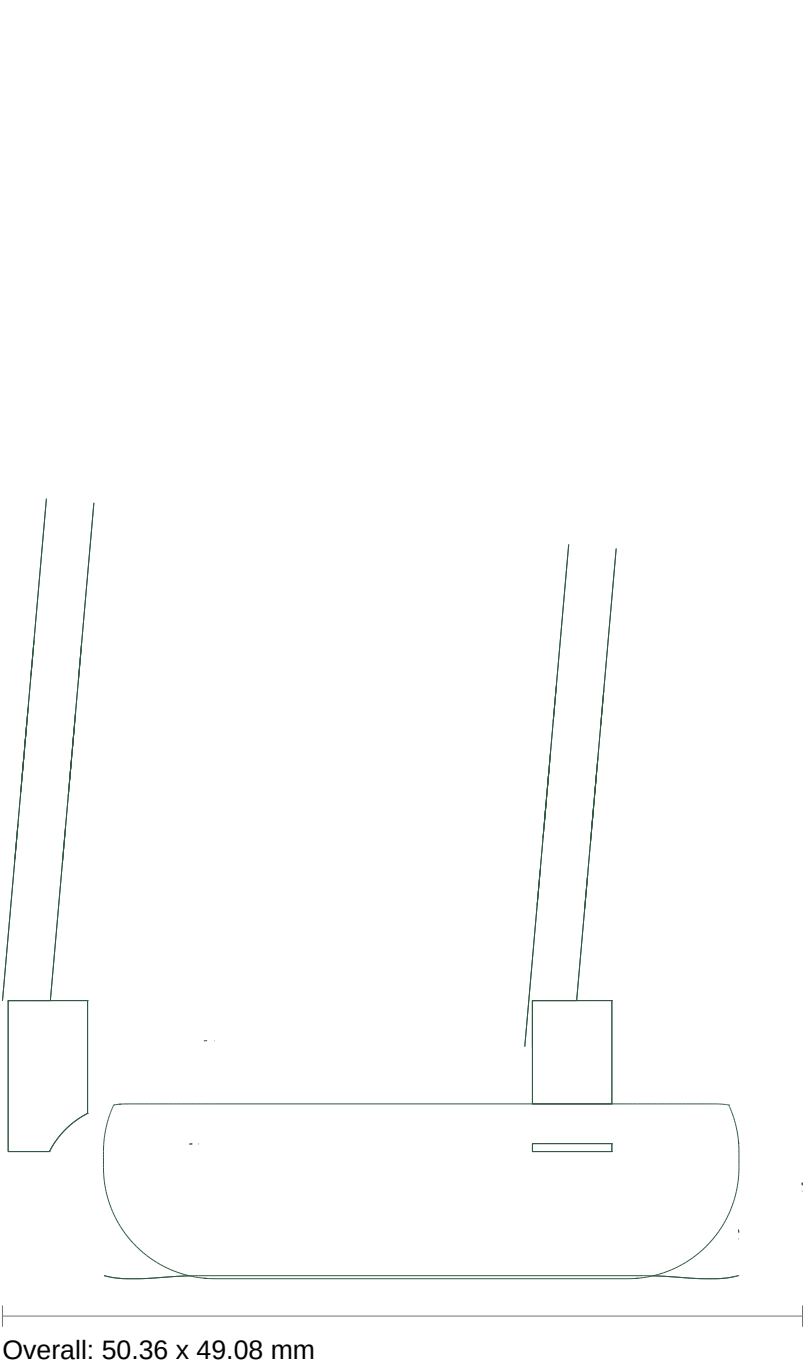
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 90.0 / 50.36 / 49.08 mm

Material volume: 28587.1 mm3 | PLA solid-equivalent: 35.45 g (slicer mass will be lower)

Foot: ankle horn/idler plates on a wide rounded sole with a toe bumper; replaceable TPU sole pad is a follow-up.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

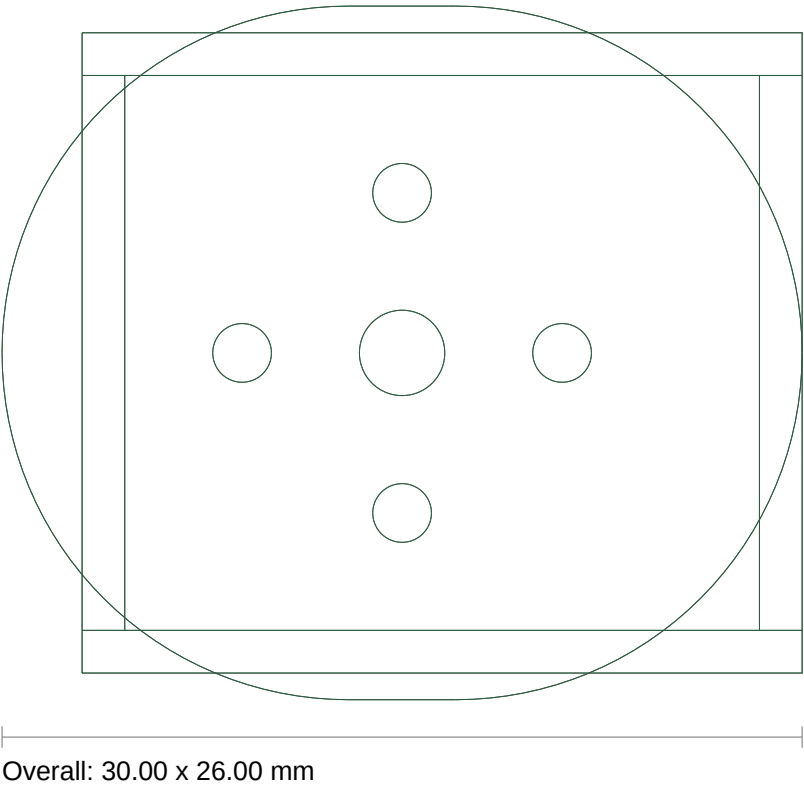
MICRO X Rev A / right_yaw2roll

link: right_yaw2roll | millimetres | original design, prototype reference, not production released

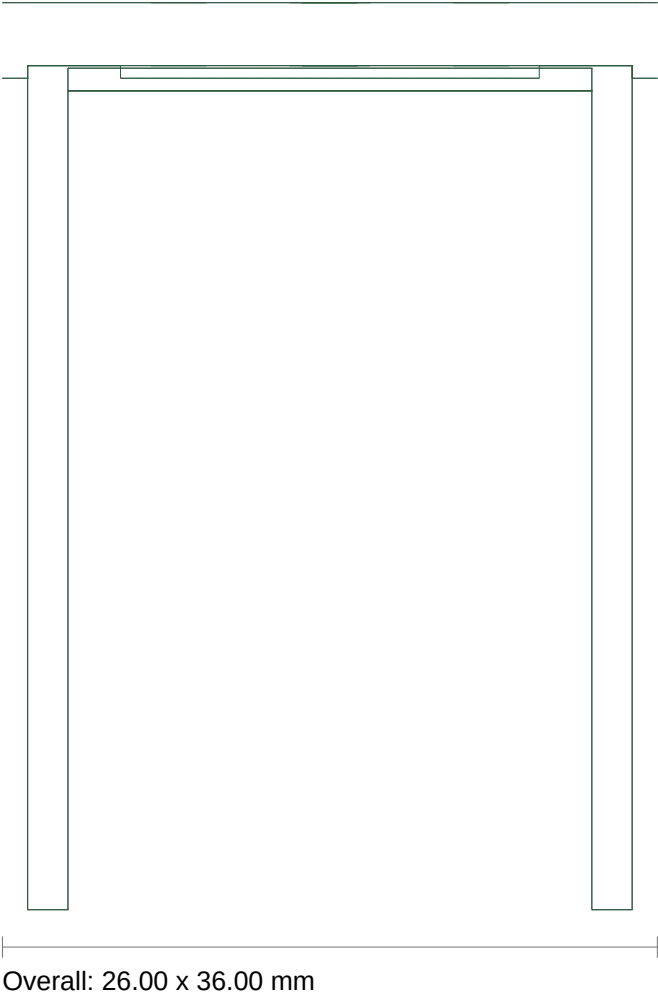
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 30.0 / 26.0 / 36.0 mm

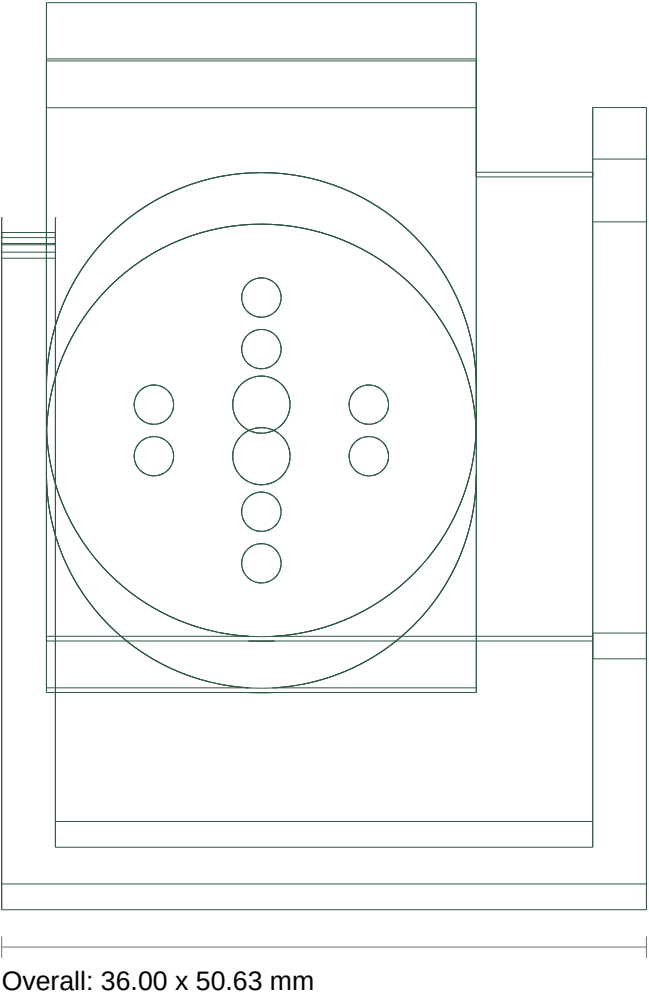
Material volume: 4568.0 mm3 | PLA solid-equivalent: 5.66 g (slicer mass will be lower)

Hip yaw horn plate with hip roll servo cradle; servo drops in from below and is retained by the hip bracket horn/idler bolts. M2 case screws to be confirmed.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

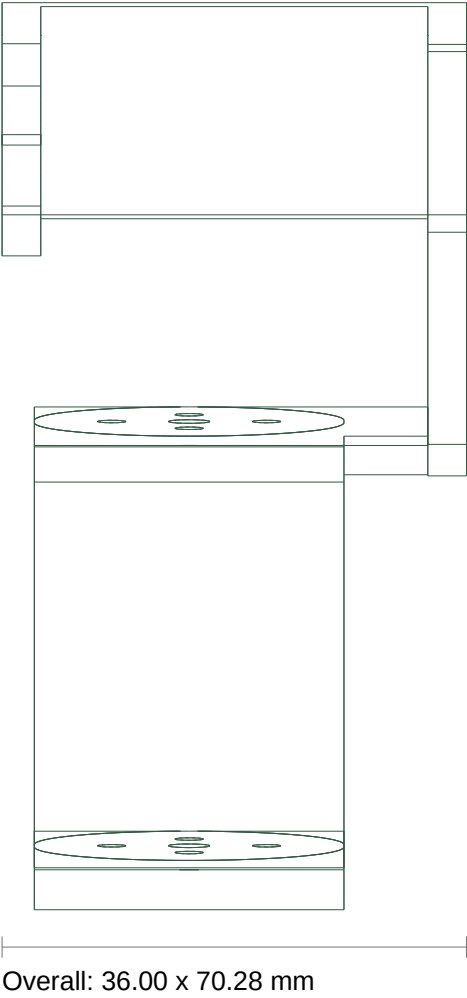
MICRO X Rev A / right_hip

link: right_hip | millimetres | original design, prototype reference, not production released

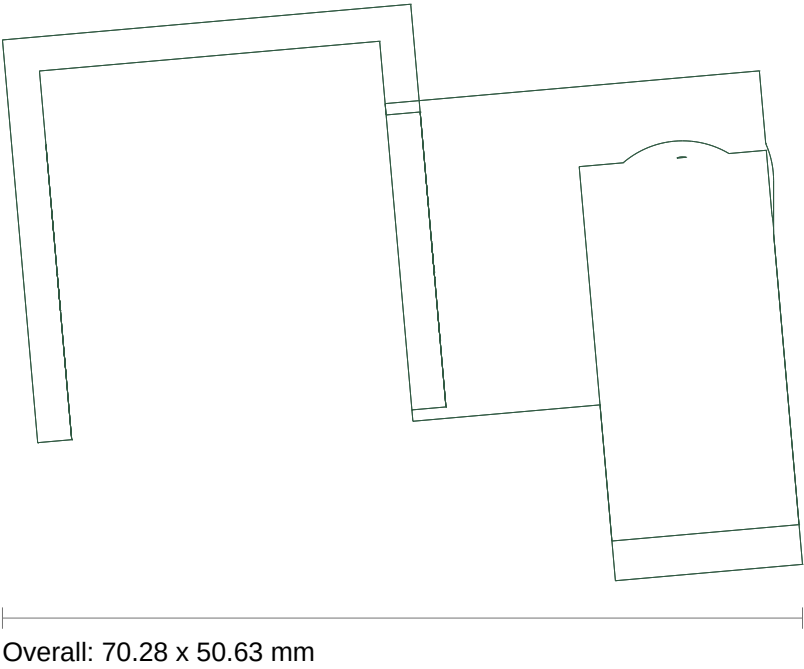
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 36.0 / 70.28 / 50.63 mm

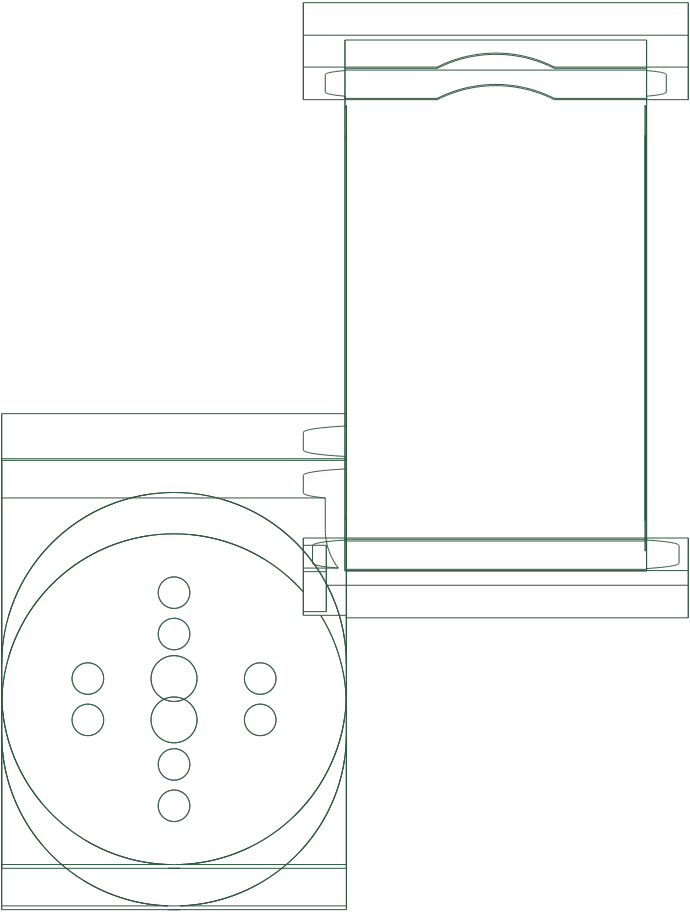
Material volume: 14895.1 mm3 | PLA solid-equivalent: 18.47 g (slicer mass will be lower)

Hip bracket: roll horn/idler plates with an under-bar, pitch horn/idler plates and a tie above the pitch servo cap. Pitch servo body lives in the thigh. Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / right_thigh

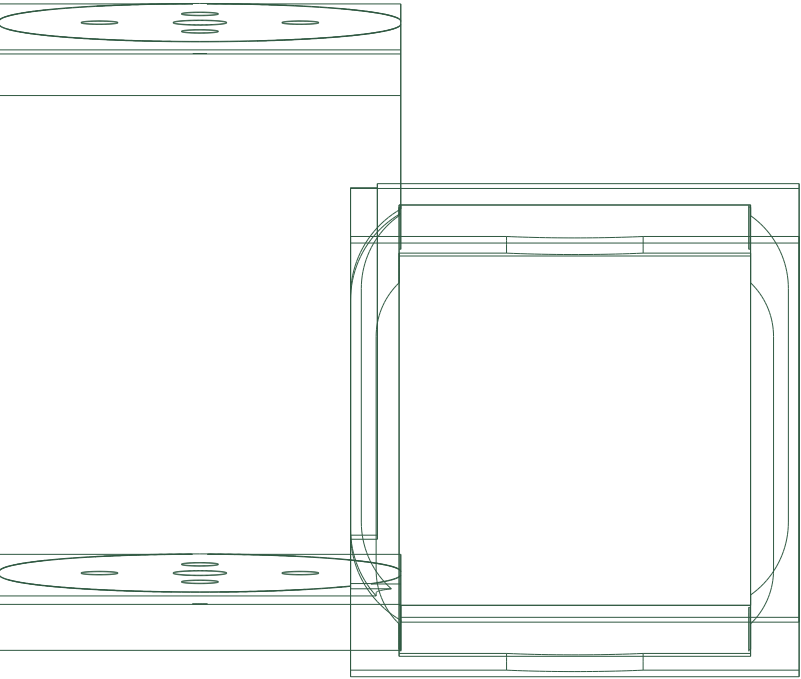
link: right_thigh | millimetres | original design, prototype reference, not production released

SIDE X-Z



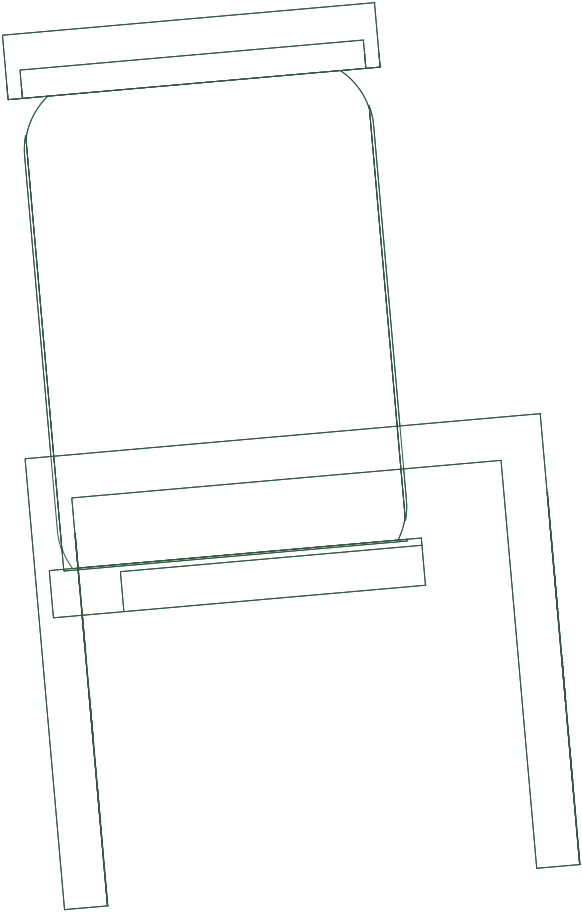
Overall: 47.80 x 63.13 mm

TOP X-Y



Overall: 47.80 x 40.21 mm

FRONT Y-Z



Overall: 40.21 x 63.13 mm

Assembly-frame extents X/Y/Z: 47.8 / 40.21 / 63.13 mm

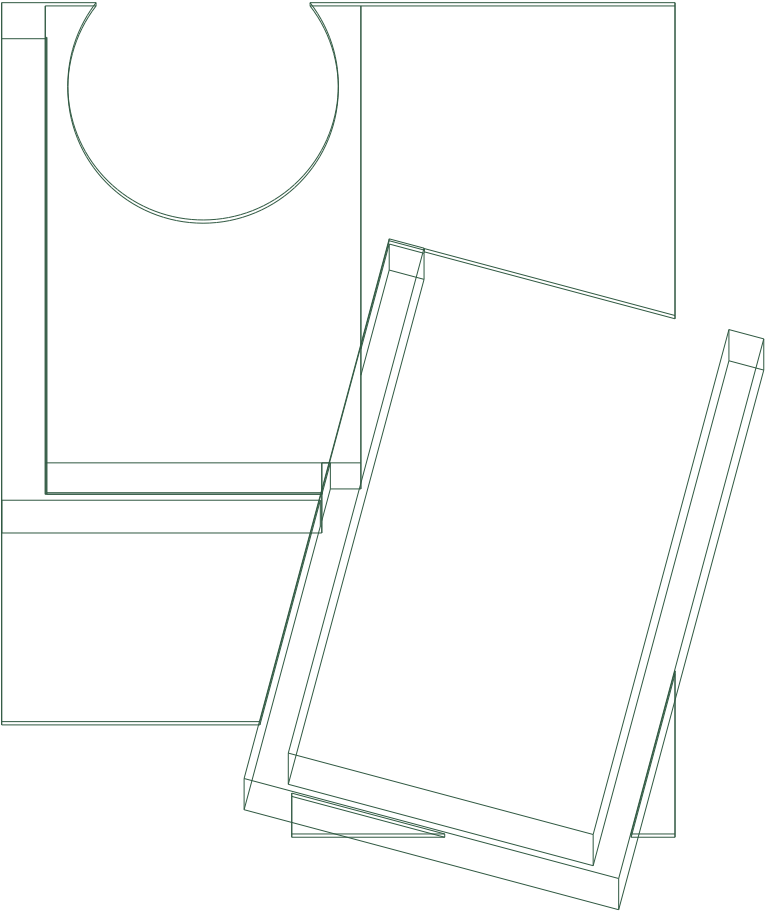
Material volume: 14711.7 mm3 | PLA solid-equivalent: 18.24 g (slicer mass will be lower)

Thigh: houses the hip pitch servo, carries the knee horn and idler plates with a tie above the knee servo.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / right_shin

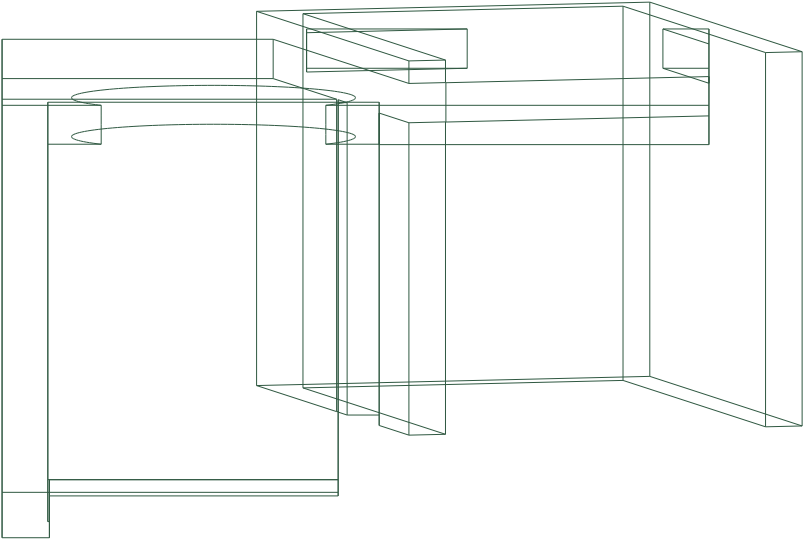
link: right_shin | millimetres | original design, prototype reference, not production released

SIDE X-Z



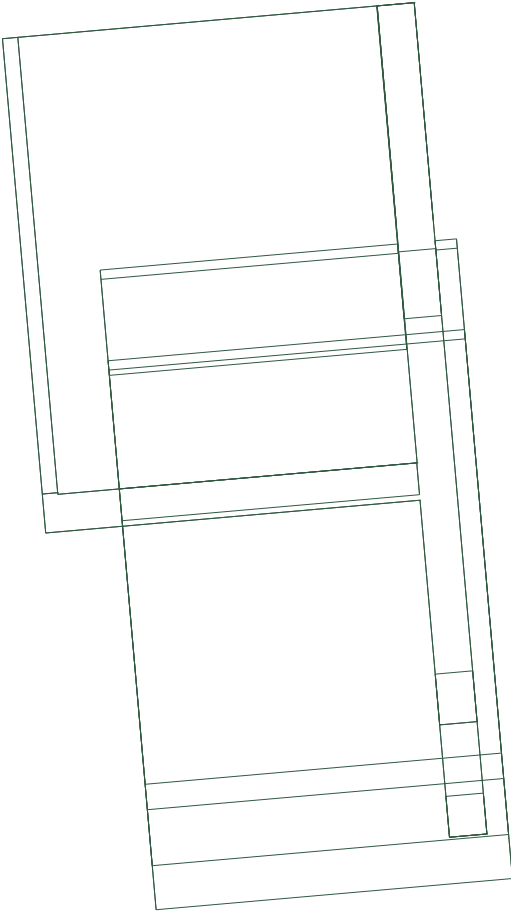
Overall: 50.71 x 60.33 mm

TOP X-Y



Overall: 50.71 x 33.93 mm

FRONT Y-Z



Overall: 33.93 x 60.33 mm

Assembly-frame extents X/Y/Z: 50.71 / 33.93 / 60.33 mm

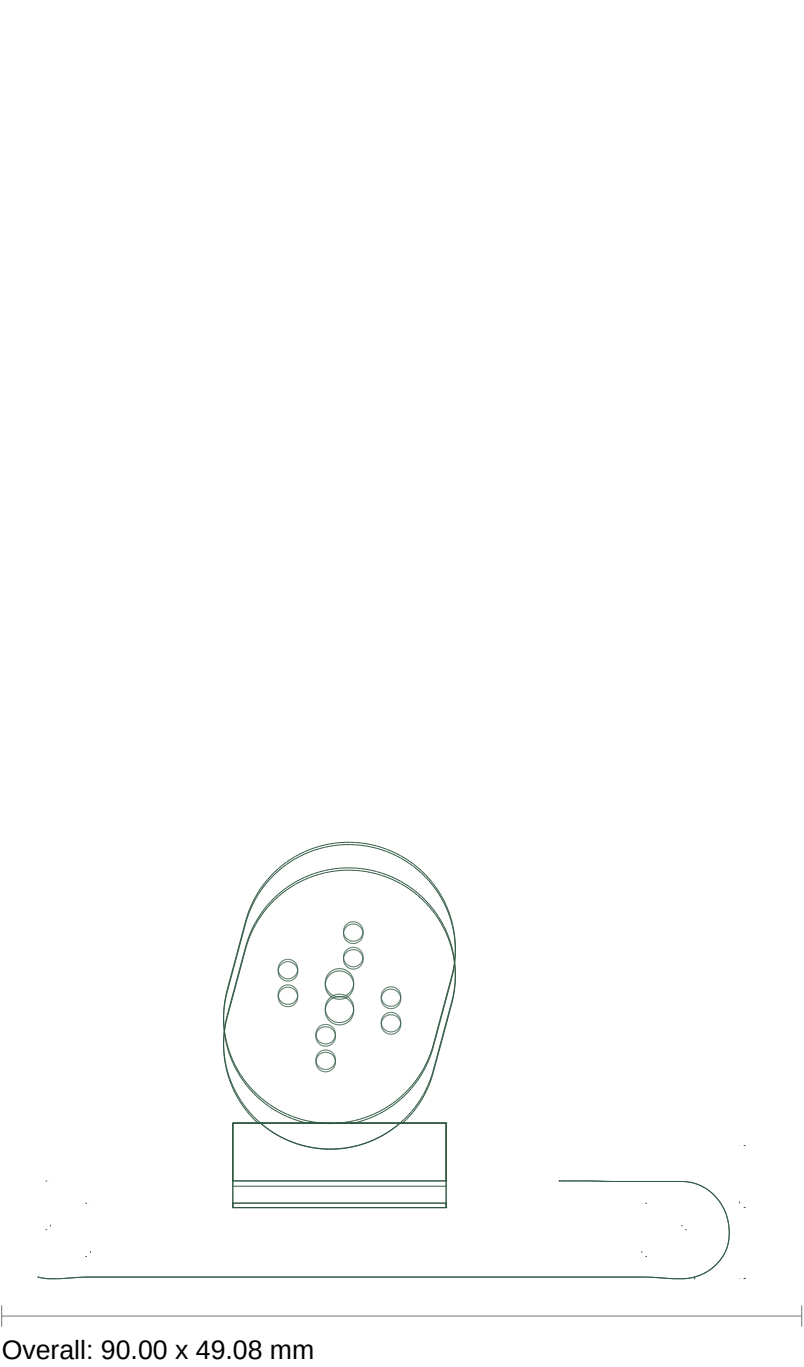
Material volume: 11986.8 mm3 | PLA solid-equivalent: 14.86 g (slicer mass will be lower)

Shin: spine plate with knee servo bay above an upright ankle servo cradle; both servos retained by the neighbouring clevis plates.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

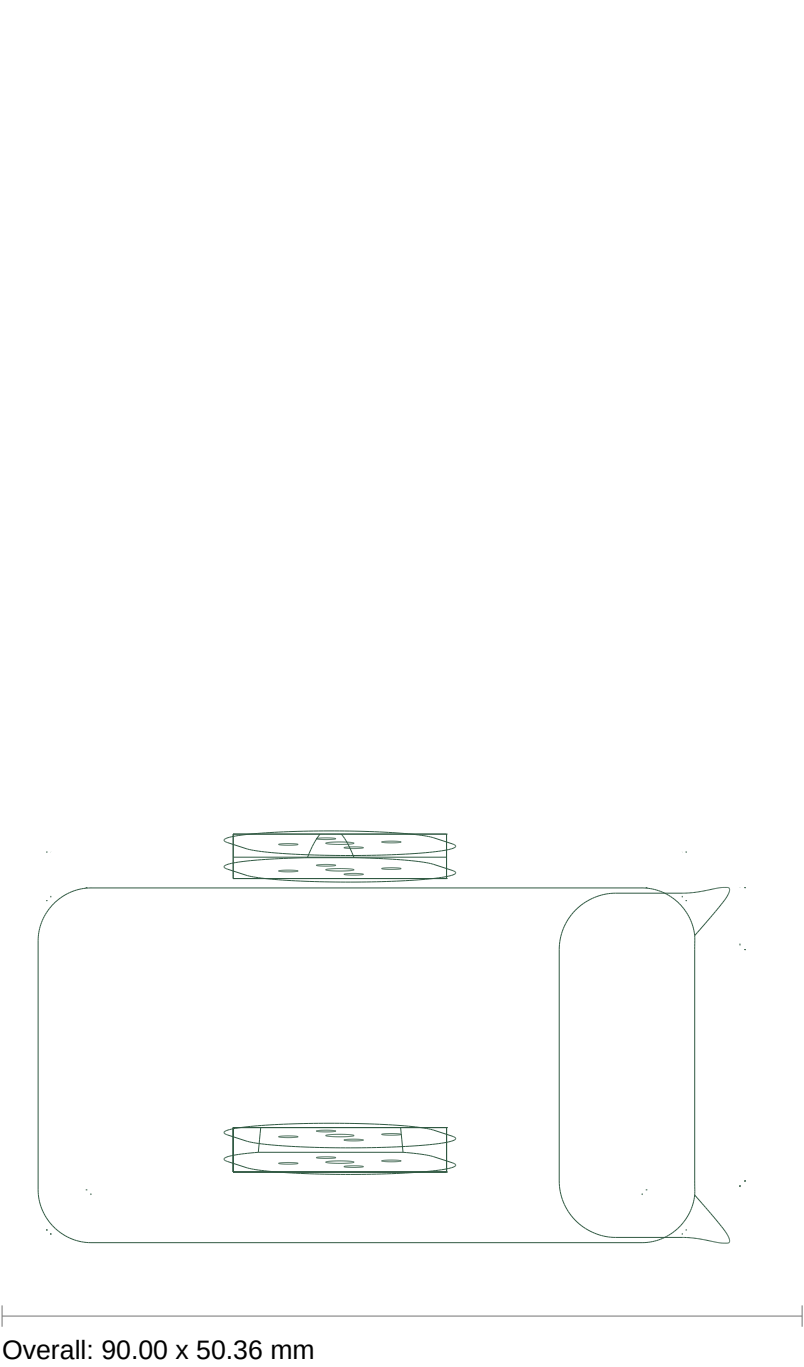
MICRO X Rev A / right_foot

link: right_foot | millimetres | original design, prototype reference, not production released

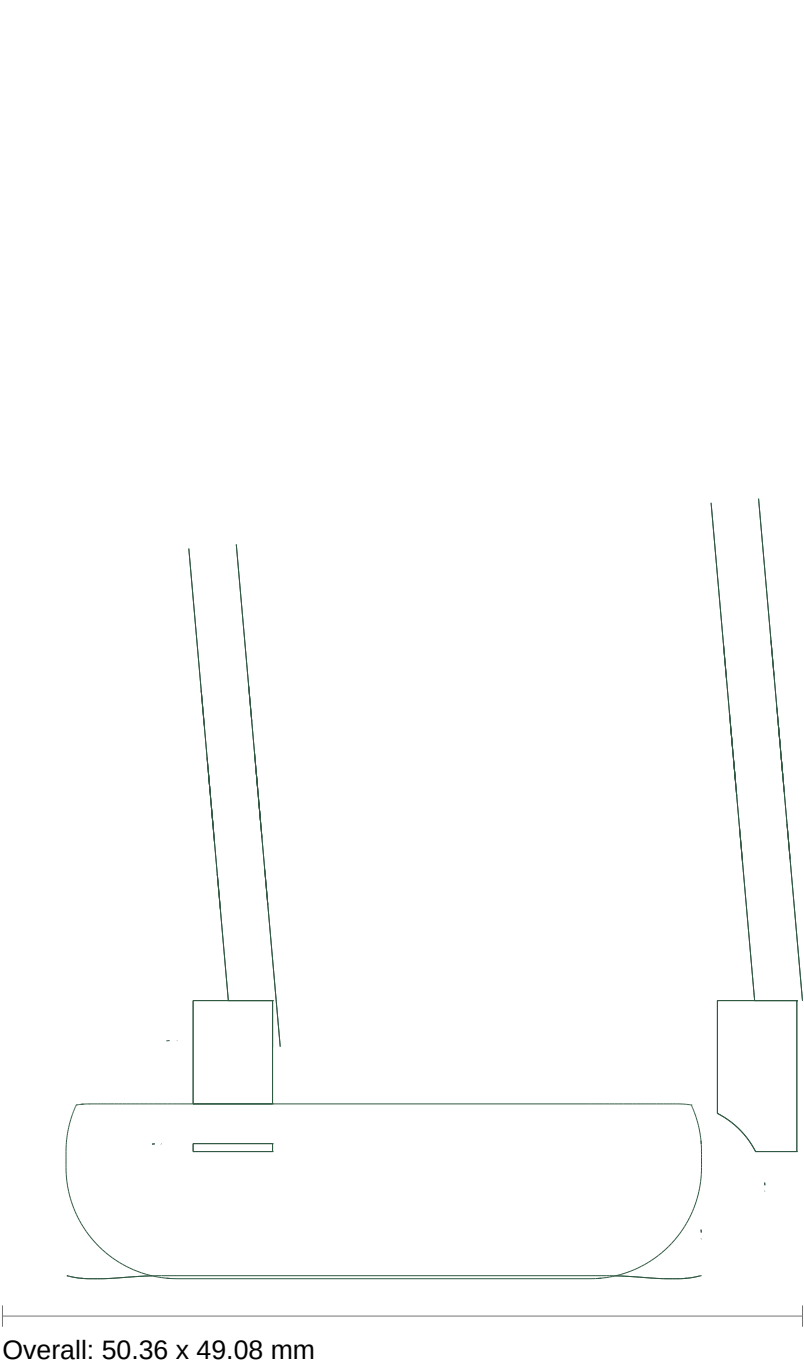
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 90.0 / 50.36 / 49.08 mm

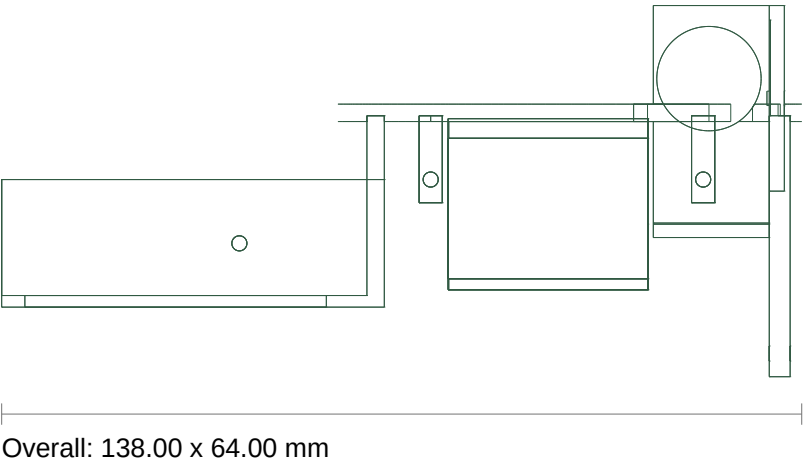
Material volume: 28587.1 mm3 | PLA solid-equivalent: 35.45 g (slicer mass will be lower)

Foot: ankle horn/idler plates on a wide rounded sole with a toe bumper; replaceable TPU sole pad is a follow-up.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

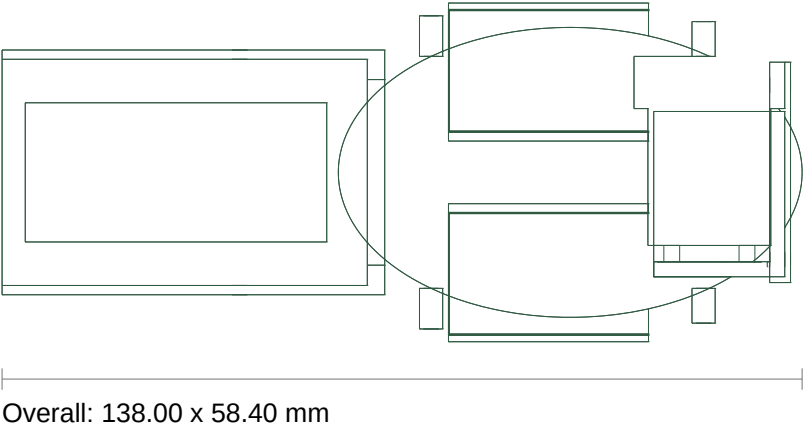
MICRO X Rev A / chassis

link: trunk | millimetres | original design, prototype reference, not production released

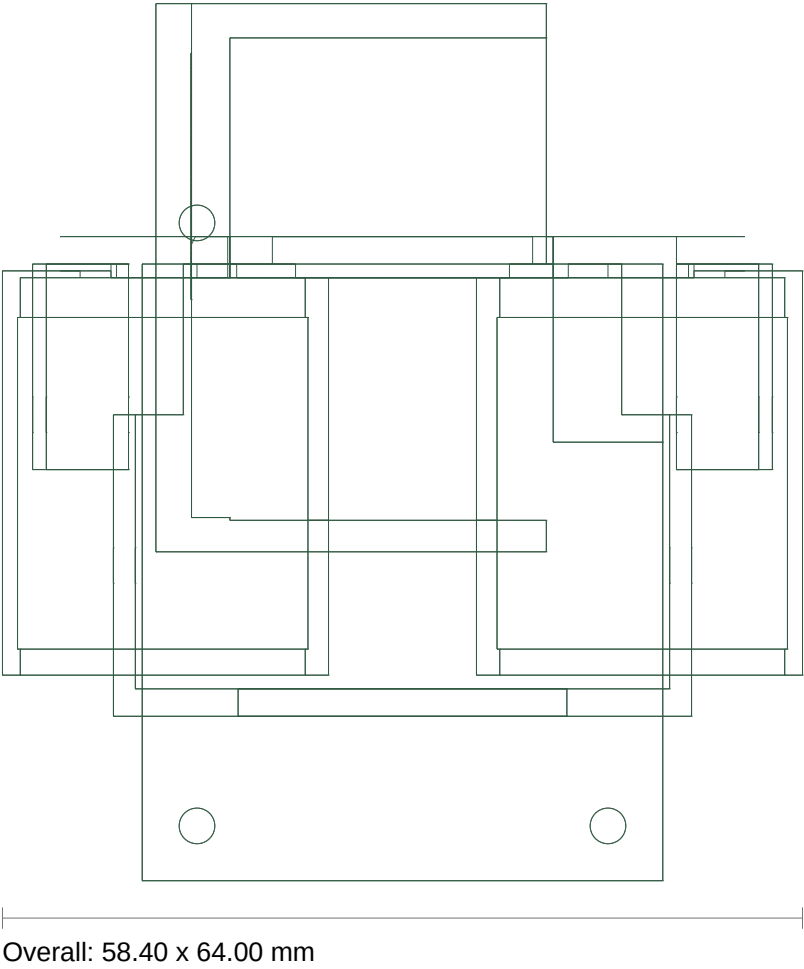
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 138.0 / 58.4 / 64.0 mm

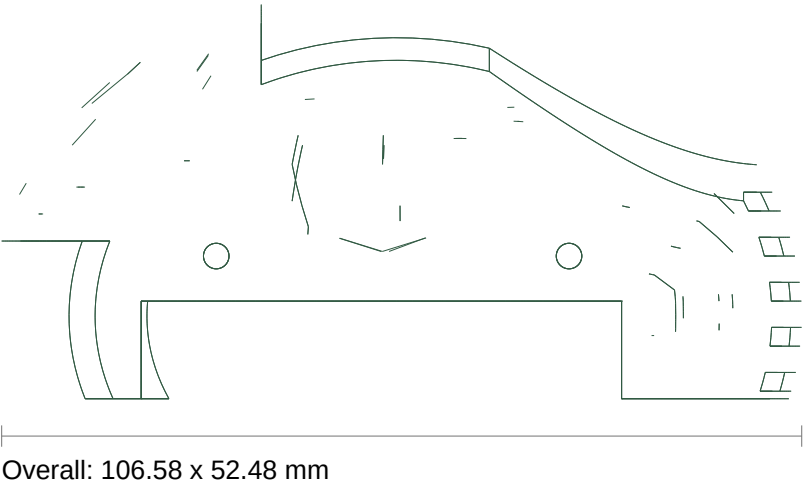
Material volume: 34953.8 mm3 | PLA solid-equivalent: 43.34 g (slicer mass will be lower)

Internal chassis: hip yaw hangers, neck servo bay, battery tray (NP-F550 class), chest board wall and four M3 shell bosses.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

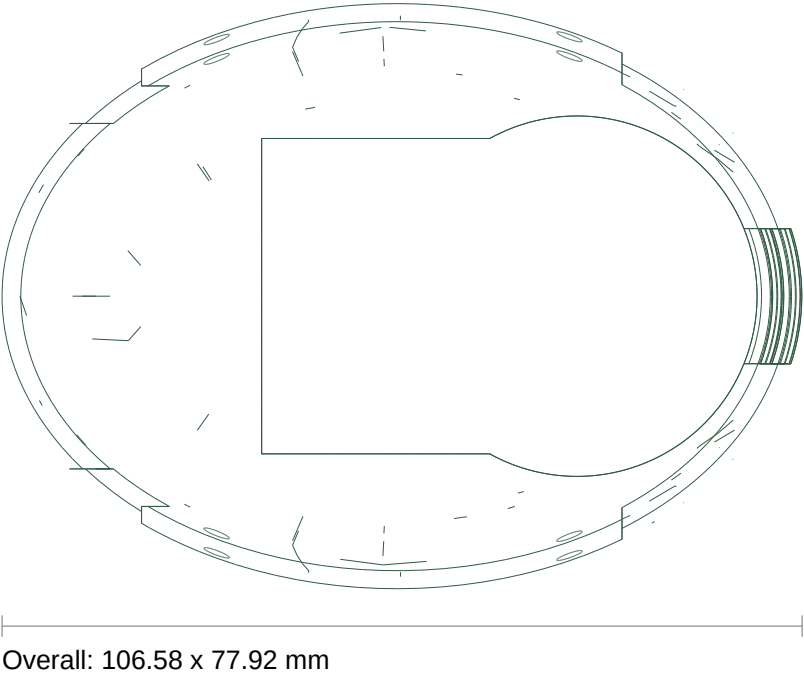
MICRO X Rev A / torso_shell

link: trunk | millimetres | original design, prototype reference, not production released

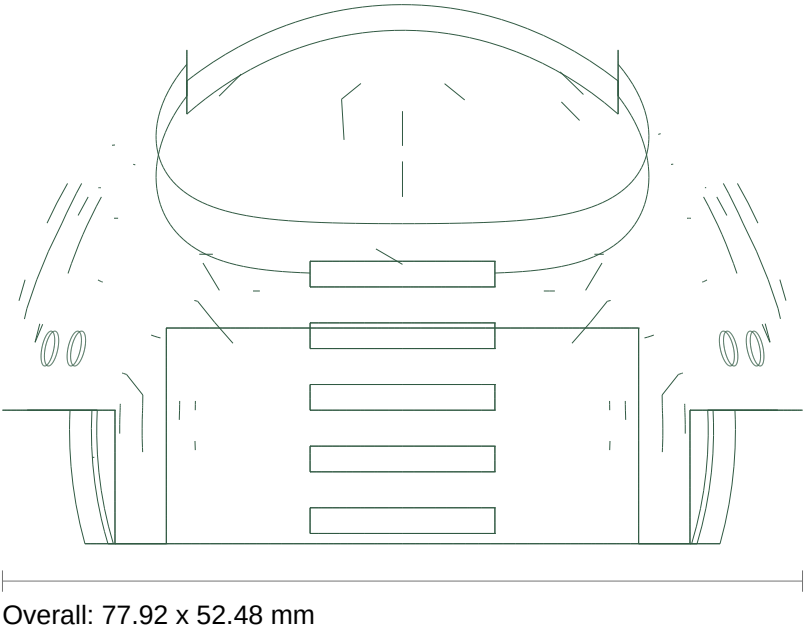
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 106.58 / 77.92 / 52.48 mm

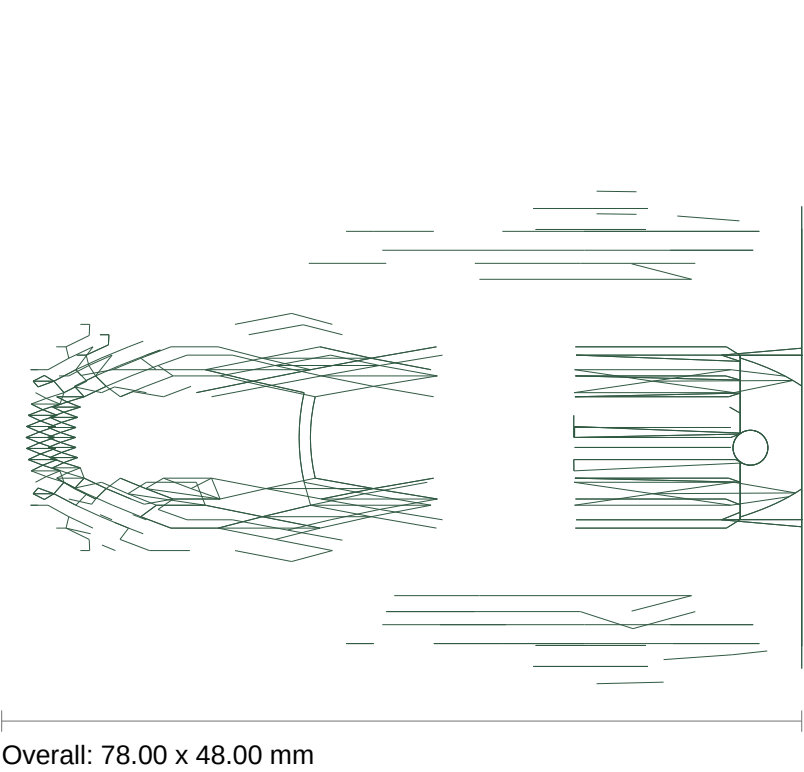
Material volume: 21619.0 mm3 | PLA solid-equivalent: 26.81 g (slicer mass will be lower)

One-piece egg shell, open below the hips, with neck and tail-battery openings and a chest grille. Four M3 through the flanks into chassis bosses; remove it to reach the board.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

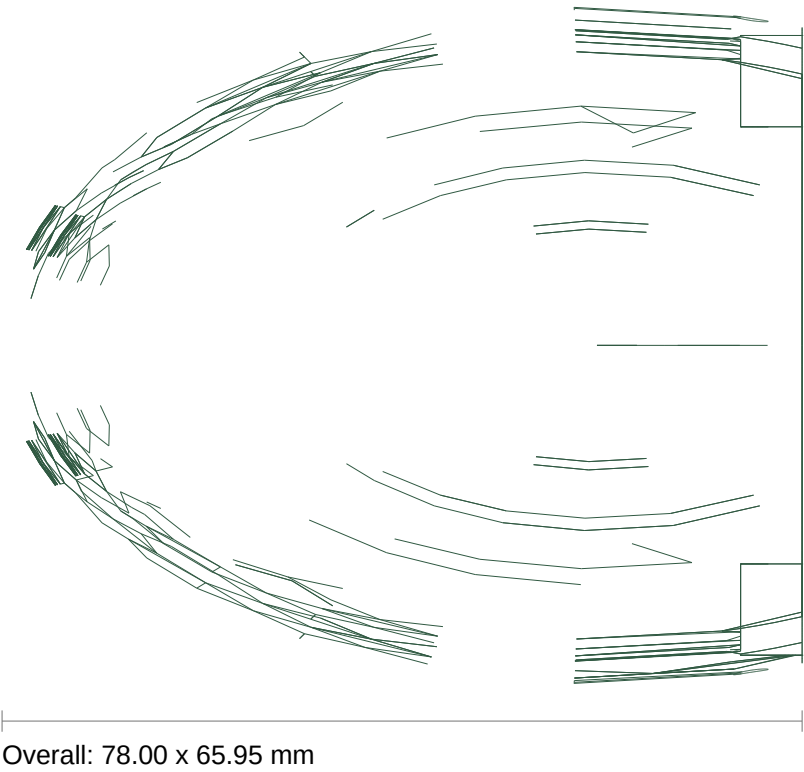
MICRO X Rev A / tail_cover

link: trunk | millimetres | original design, prototype reference, not production released

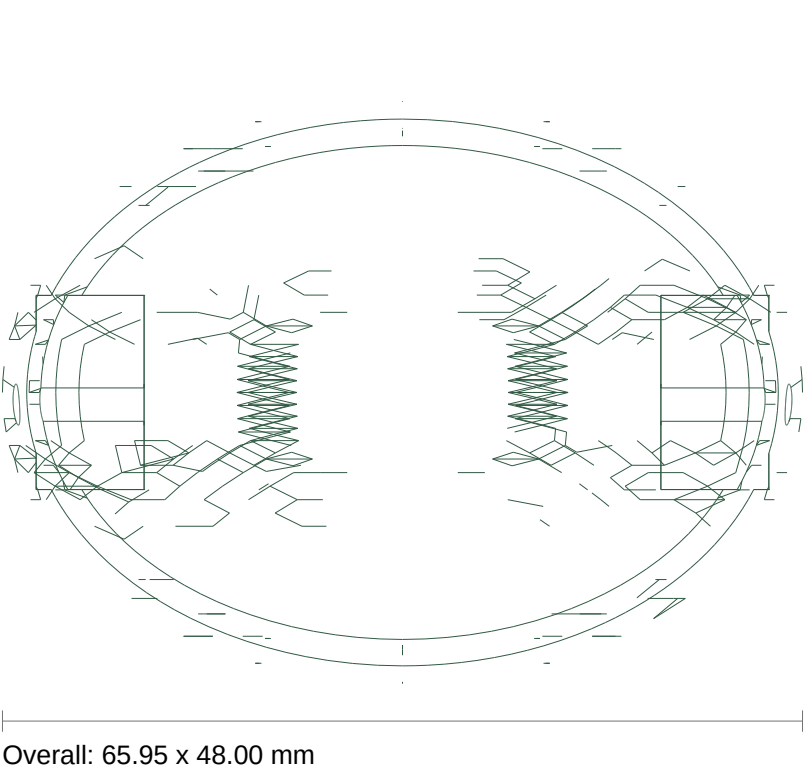
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 78.0 / 65.95 / 48.0 mm

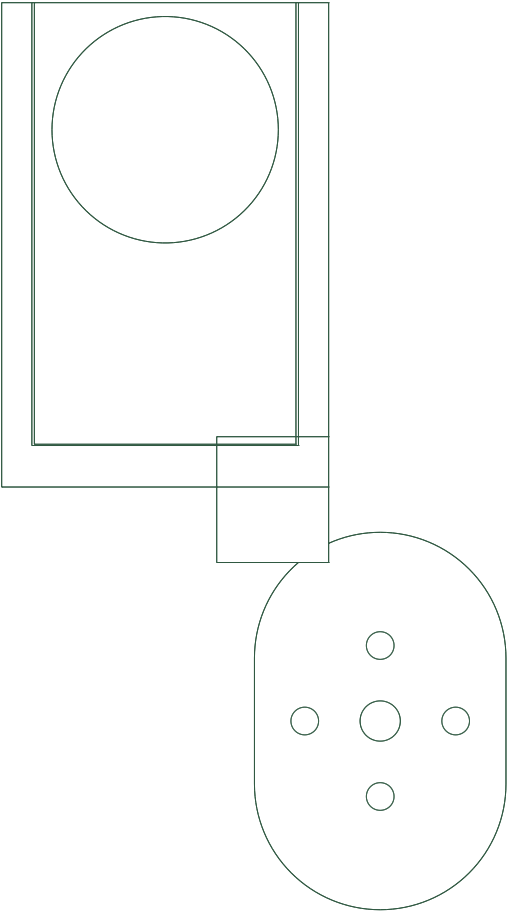
Material volume: 26391.1 mm3 | PLA solid-equivalent: 32.72 g (slicer mass will be lower)

Tail = battery cover: slides over the tray and holds an NP-F550 class pack; two M3 screws into the tray.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / neck_link

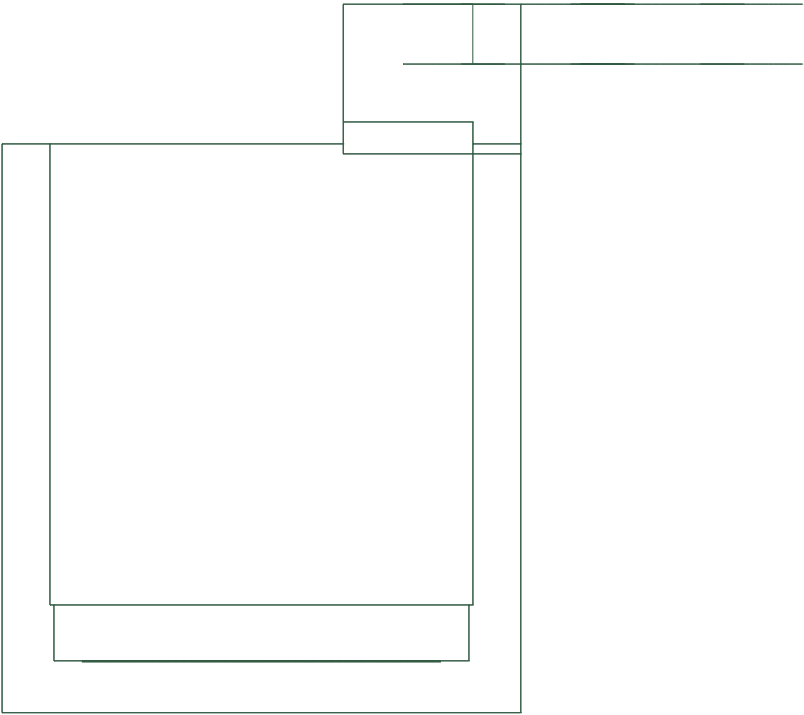
link: neck | millimetres | original design, prototype reference, not production released

SIDE X-Z



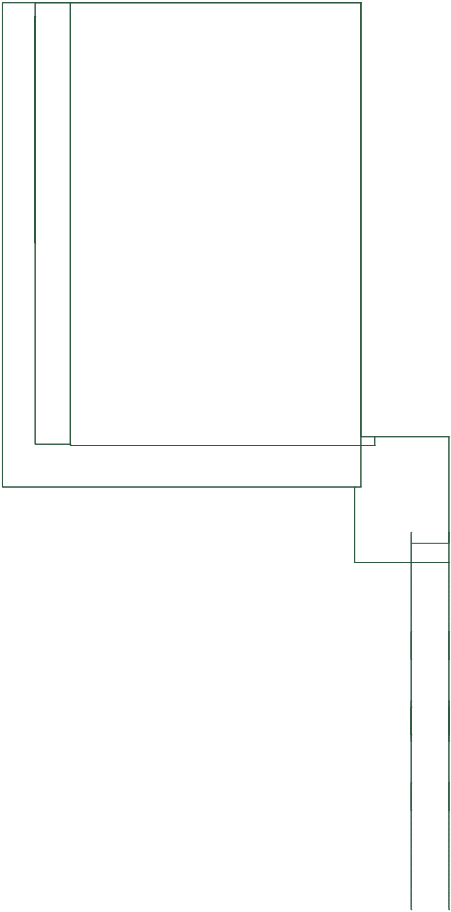
Overall: 40.10 x 72.10 mm

TOP X-Y



Overall: 40.10 x 35.50 mm

FRONT Y-Z



Overall: 35.50 x 72.10 mm

Assembly-frame extents X/Y/Z: 40.1 / 35.5 / 72.1 mm

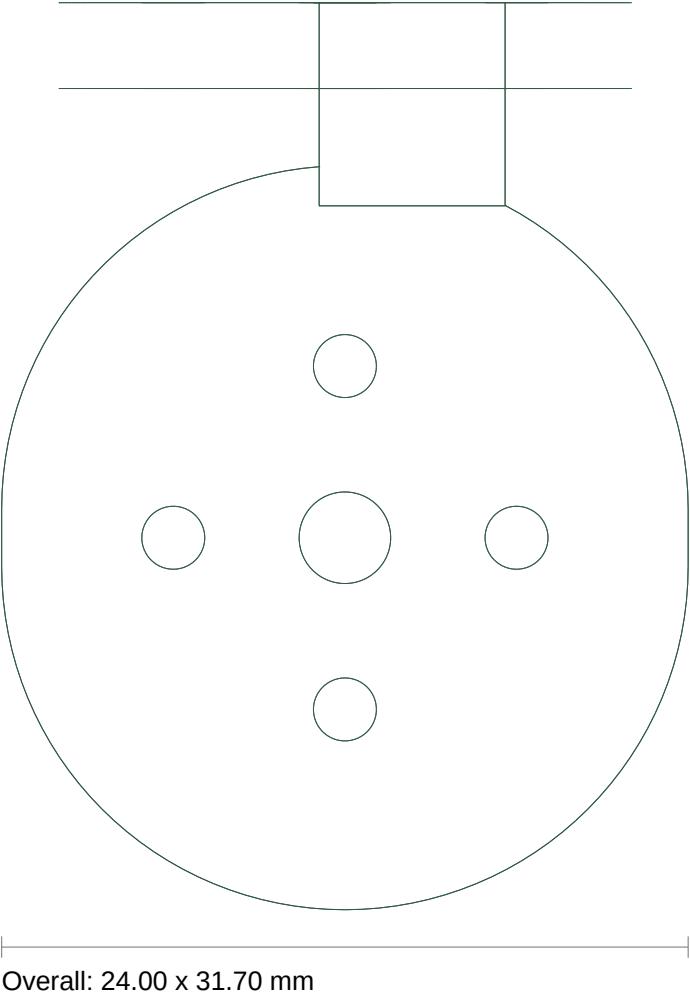
Material volume: 11345.3 mm3 | PLA solid-equivalent: 14.07 g (slicer mass will be lower)

Neck: neck-pitch horn plate rising into the head-pitch servo bay.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

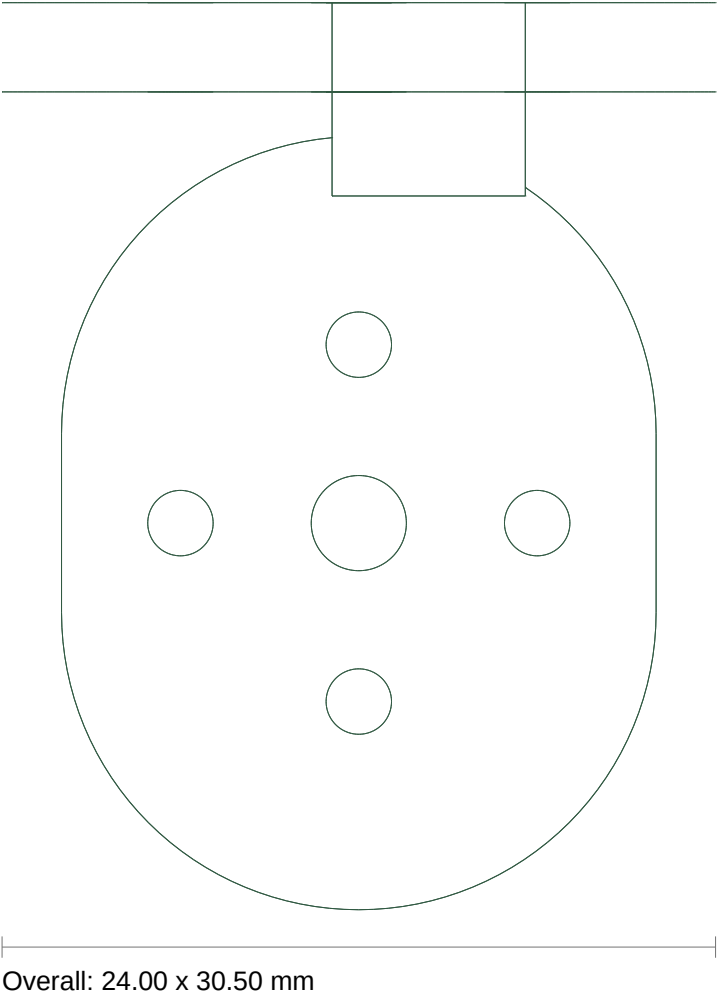
MICRO X Rev A / head_base

link: head_base | millimetres | original design, prototype reference, not production released

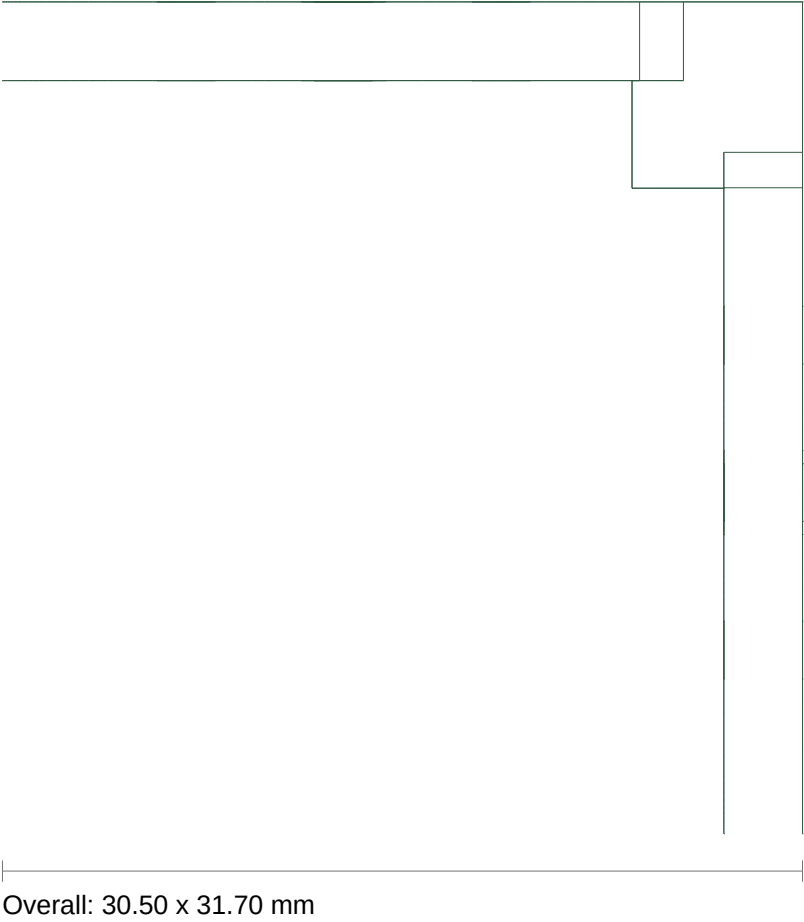
SIDE X-Z



TOP X-Y



FRONT Y-Z



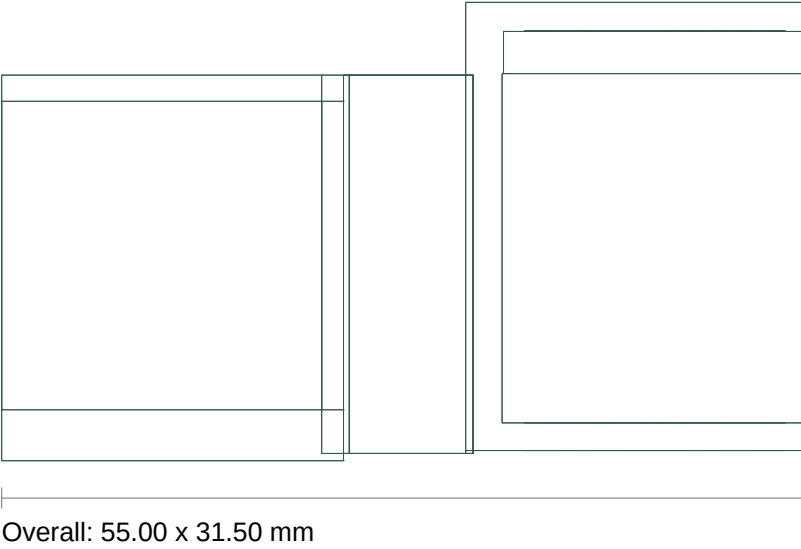
Assembly-frame extents X/Y/Z: 24.0 / 30.5 / 31.7 mm
Material volume: 2915.8 mm3 | PLA solid-equivalent: 3.62 g (slicer mass will be lower)

Head base: head-pitch horn plate and the fixed head-yaw horn plate; the only part that stays still while the head yaws.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

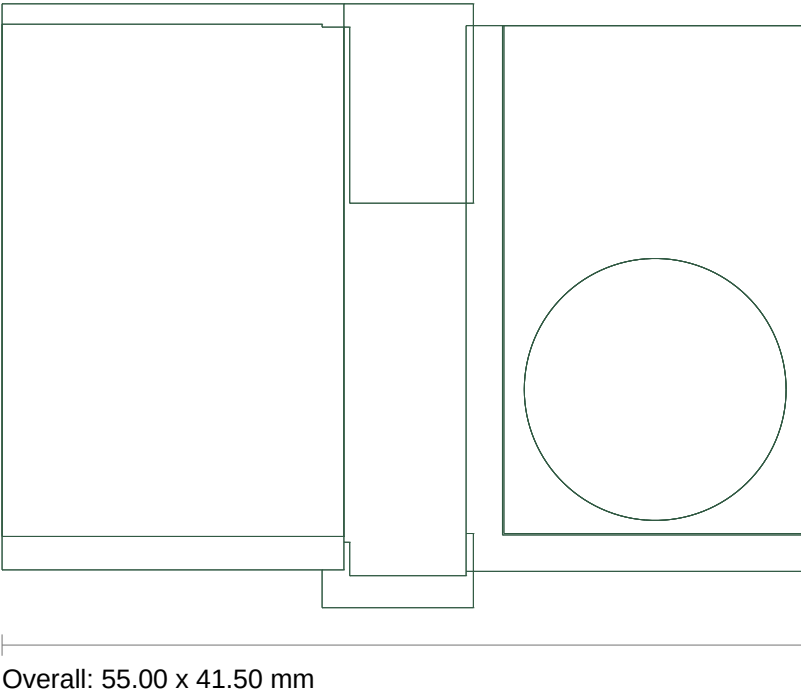
MICRO X Rev A / head_yoke

link: head_yoke | millimetres | original design, prototype reference, not production released

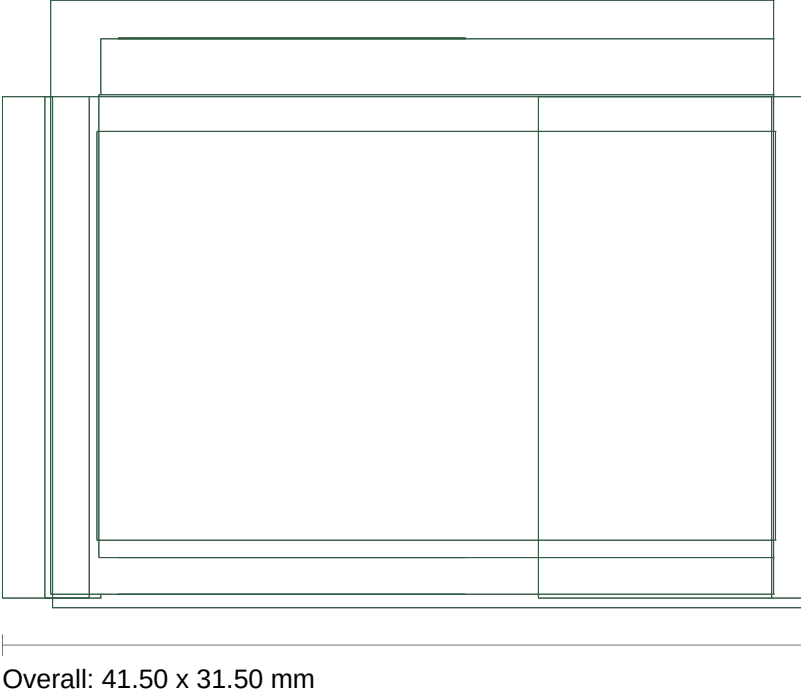
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 55.0 / 41.5 / 31.5 mm

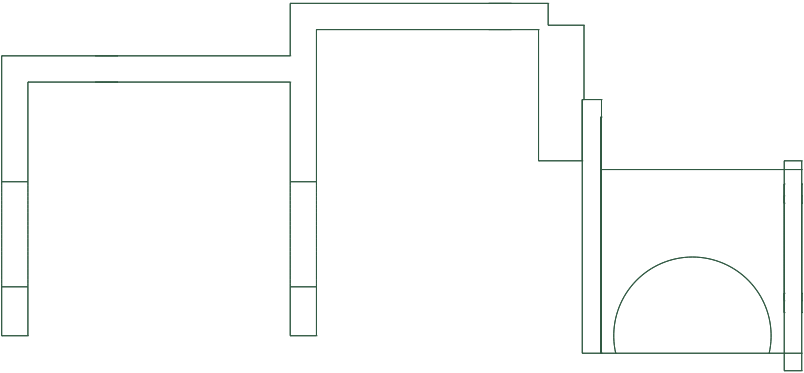
Material volume: 17025.7 mm3 | PLA solid-equivalent: 21.11 g (slicer mass will be lower)

Yoke: carries the head-yaw servo body (horn down onto the base) and the head-roll servo.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / head_frame

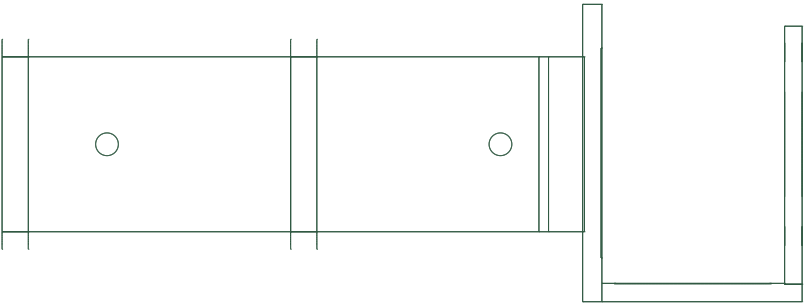
link: head | millimetres | original design, prototype reference, not production released

SIDE X-Z



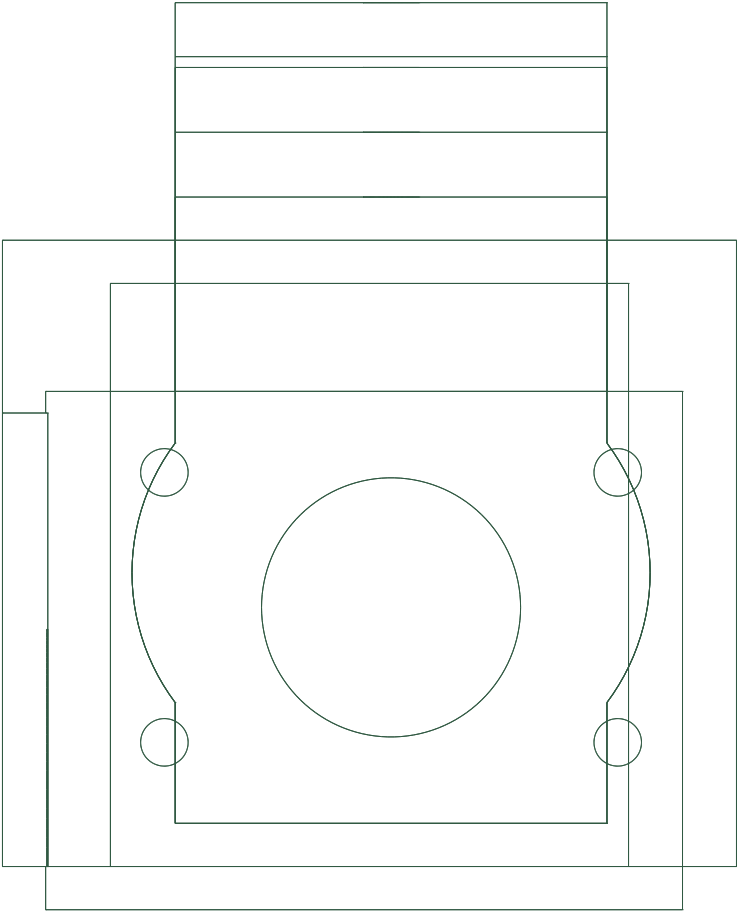
Overall: 91.50 x 42.00 mm

TOP X-Y



Overall: 91.50 x 34.00 mm

FRONT Y-Z



Overall: 34.00 x 42.00 mm

Assembly-frame extents X/Y/Z: 91.5 / 34.0 / 42.0 mm

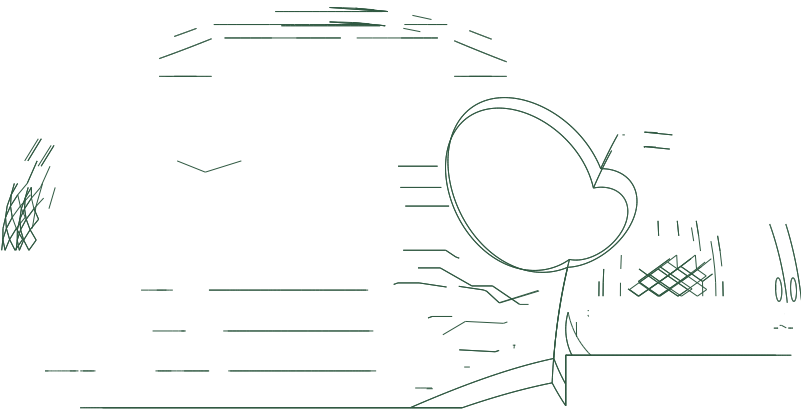
Material volume: 13589.1 mm3 | PLA solid-equivalent: 16.85 g (slicer mass will be lower)

Head frame: head-roll clevis, top rail, jaw servo bay and the camera carrier wall (Camera Module 3 hole pattern). Skull screws to the rail. Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / skull

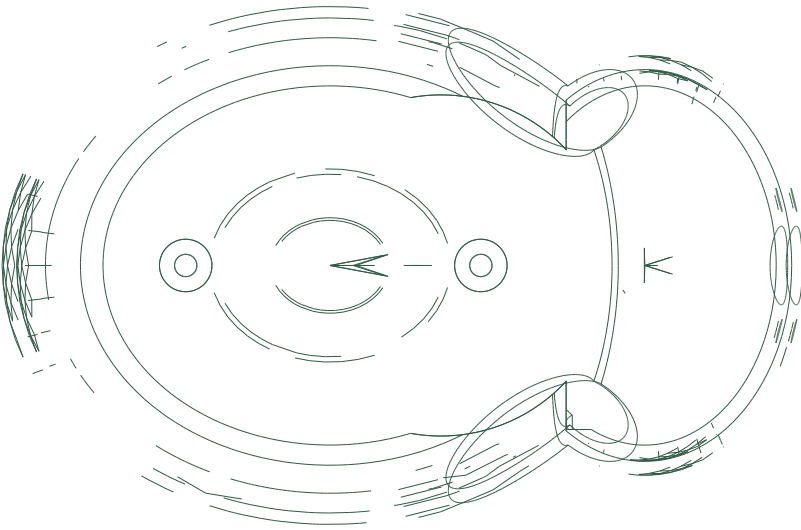
link: head | millimetres | original design, prototype reference, not production released

SIDE X-Z



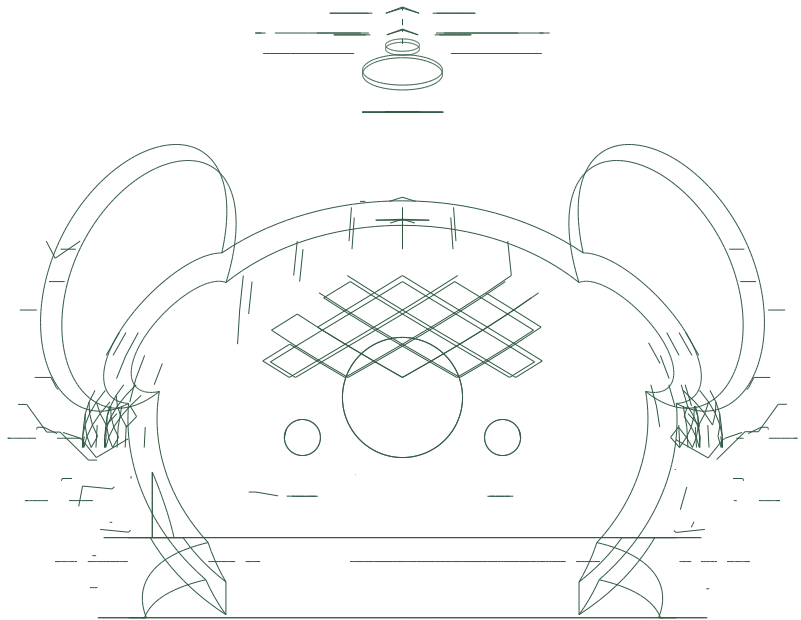
Overall: 121.98 x 61.00 mm

TOP X-Y



Overall: 121.98 x 79.94 mm

FRONT Y-Z



Overall: 79.94 x 61.00 mm

Assembly-frame extents X/Y/Z: 121.98 / 79.94 / 61.0 mm

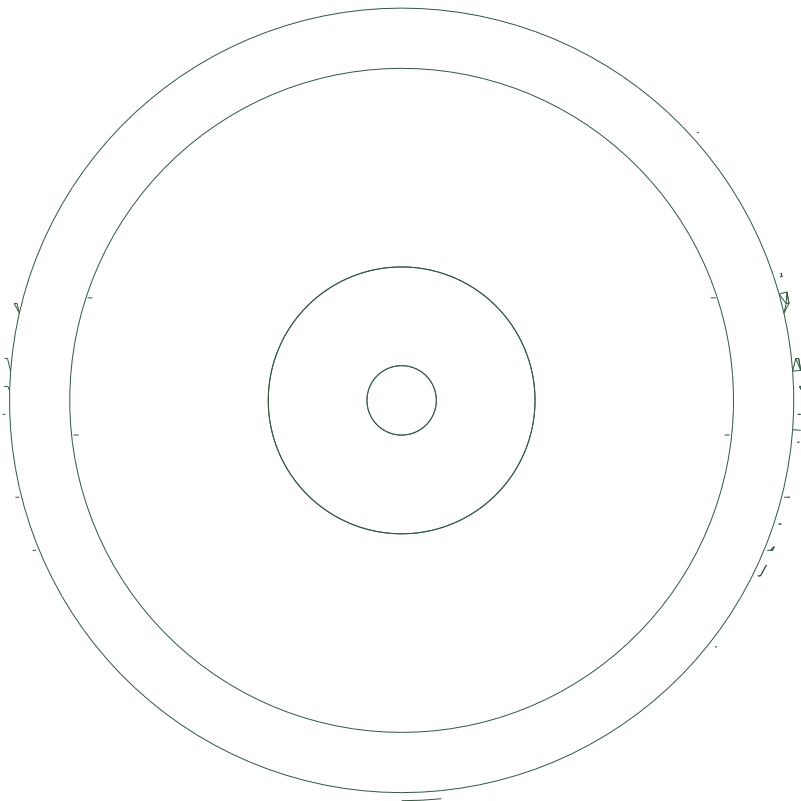
Material volume: 38623.5 mm3 | PLA solid-equivalent: 47.89 g (slicer mass will be lower)

Skull: one-piece hollow head with muzzle, eye sockets, nose camera port and nostrils. Two M3 screws down into the head frame rail. Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / eye_left

link: head | millimetres | original design, prototype reference, not production released

SIDE X-Z



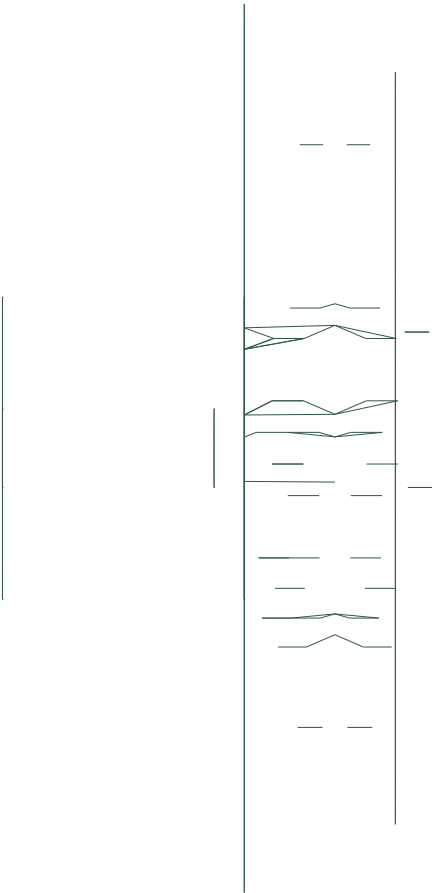
Overall: 29.99 x 30.00 mm

TOP X-Y



Overall: 29.99 x 25.99 mm

FRONT Y-Z



Overall: 25.99 x 30.00 mm

Assembly-frame extents X/Y/Z: 29.99 / 25.99 / 30.0 mm

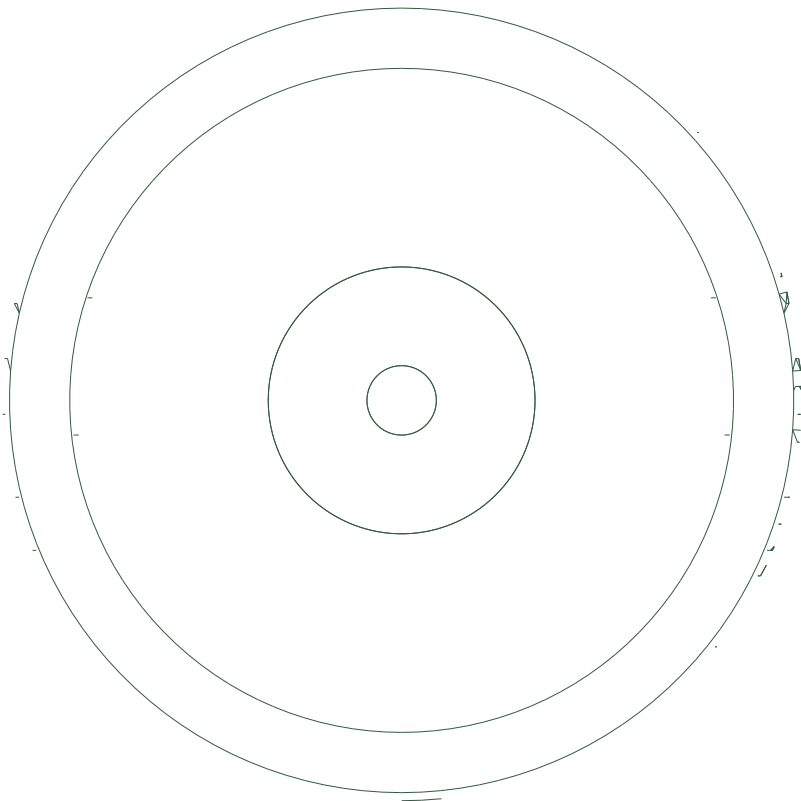
Material volume: 6551.7 mm3 | PLA solid-equivalent: 8.12 g (slicer mass will be lower)

Eye cap: Ø28 dome with Ø10 stem, fixed from inside the skull with an M3 plastic screw. Pupil and highlight are paint intent.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

MICRO X Rev A / eye_right

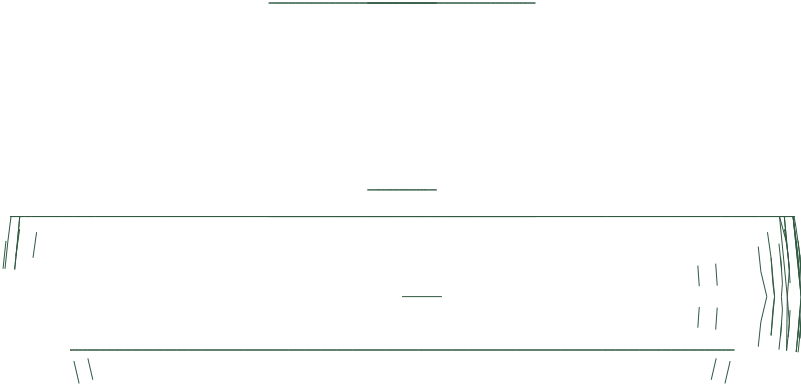
link: head | millimetres | original design, prototype reference, not production released

SIDE X-Z



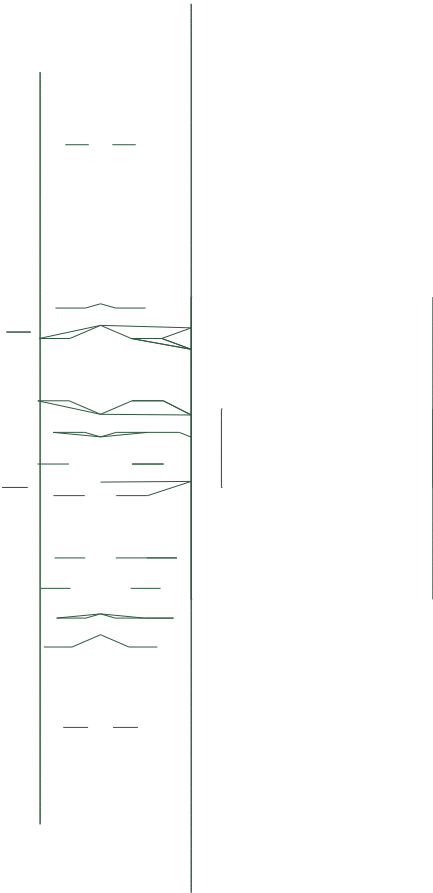
Overall: 29.99 x 30.00 mm

TOP X-Y



Overall: 29.99 x 25.99 mm

FRONT Y-Z



Overall: 25.99 x 30.00 mm

Assembly-frame extents X/Y/Z: 29.99 / 25.99 / 30.0 mm

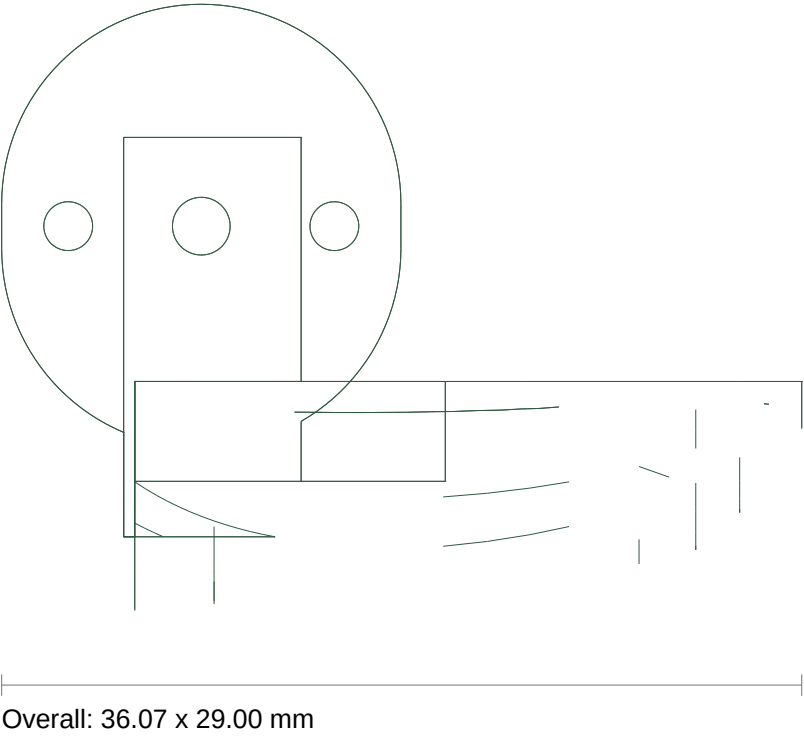
Material volume: 6551.7 mm3 | PLA solid-equivalent: 8.12 g (slicer mass will be lower)

Eye cap: Ø28 dome with Ø10 stem, fixed from inside the skull with an M3 plastic screw. Pupil and highlight are paint intent.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.

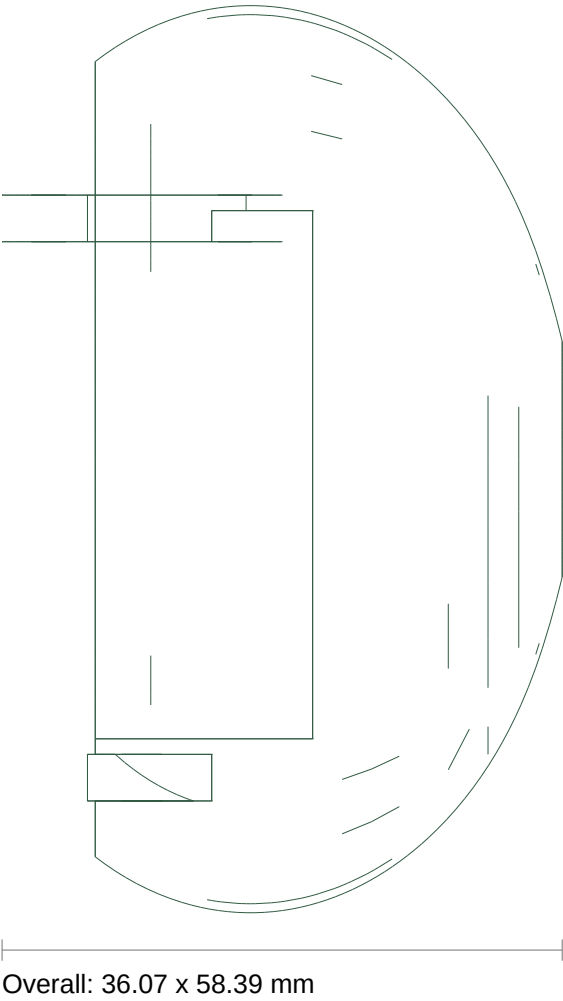
MICRO X Rev A / jaw_beak

link: jaw | millimetres | original design, prototype reference, not production released

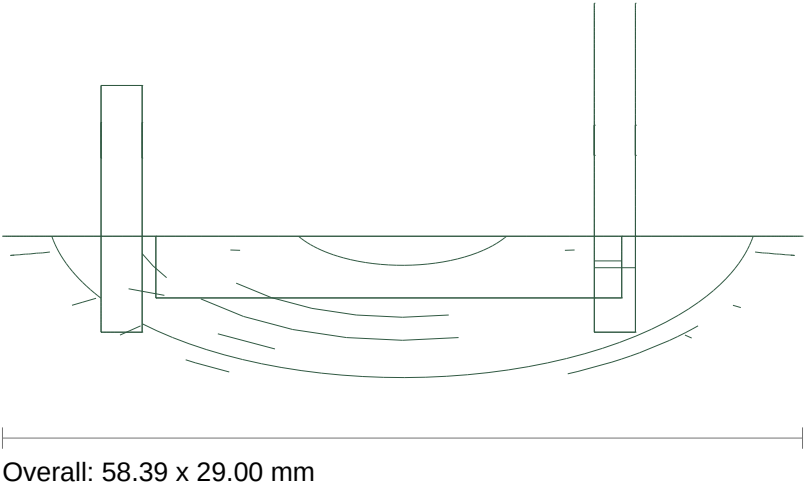
SIDE X-Z



TOP X-Y



FRONT Y-Z



Assembly-frame extents X/Y/Z: 36.07 / 58.39 / 29.0 mm

Material volume: 10984.1 mm3 | PLA solid-equivalent: 13.62 g (slicer mass will be lower)

Lower beak: solid rounded scoop with a left arm bolted to the jaw servo horn and a right pin pivot; rear trimmed to clear the neck stack when the head yaws.
Use STEP for exact geometry. Clearance, fastening, interference and physical fit require prototype verification. Rights: root LICENSE.